



ACQLERATE

MODULE 7

The Business of Defense Contracting

The MBA layer nobody teaches PMs. How your program's numbers become the company's revenue, profit, cash, backlog, and valuation. Plain English, real math, contractor-side.

11 Lessons · ~281 Minutes · Lesson Book Edition

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THE BUSINESS OF DEFENSE CONTRACTING · LESSON 1

Revenue Recognition II: How Your EAC Becomes the Company's Revenue

28 min

The plain-English version of ASC 606 for defense contractors. Why most DoD contracts book revenue "over time," how the cost-to-cost method turns your monthly EAC into the revenue number the CFO reports, and why a PM who gets the EAC wrong is quietly misstating the company's books.

The Number You Report Every Month Is Also the Company's Revenue

In the Finance module you learned to build an Estimate at Completion and to defend it. Here is the part nobody told you: on most defense contracts, your EAC is not just a program management number. It is the input the accounting team uses to calculate how much revenue and profit the company reports this month. Change the EAC and you change the income statement. That is not a metaphor. It is literally how the math works, and this lesson walks through it with real numbers so you can explain it to your team. First, a definition. Revenue recognition is the set of rules that decide when a company is allowed to record revenue. Not when it invoices, not when cash arrives, but when it has actually earned the money by doing the work. In the U.S., those rules are called ASC 606. Every defense contractor that produces financial statements, public or private, follows them.

REVENUE, BILLING, AND CASH ARE THREE DIFFERENT WORDS

Revenue, billing, and cash are three different things. You can recognize revenue on work you have not invoiced yet. You can invoice for work you recognized last month. You can wait 45 days for the cash. PMs who mix these up tell leadership "we billed \$2M" when the question was "what did we book?" Keep the three words separate and you will already sound sharper than most of the room.

The Five Steps, in Plain English

ASC 606 has five steps. The Revenue Recognition lesson in the Finance module introduced them. Here is what each one means when you are the PM on a DoD contract.

- **Step 1: Identify the contract.** Usually easy. The signed contract or task order. For an IDIQ, each task order is typically its own contract for accounting purposes. Unfunded options are not part of the contract until exercised.
- **Step 2: Identify the performance obligations.** What distinct things did you promise? Often one per CLIN, but not always. A development effort plus the delivered system may be one obligation because they are highly interdependent. Sustainment years may each be separate obligations.

- **Step 3: Determine the transaction price.** Contract value plus the portion of any variable amounts (award fee, incentive fee) the company believes it will probably earn. A CPAF contract with a 10% pool where you have historically earned 85% would count about 8.5% as expected fee, not 10%.
- **Step 4: Allocate the price to the obligations.** If a contract has several performance obligations, the transaction price is split across them based on what each would sell for on its own. On a single-CLIN services contract this step is trivial.
- **Step 5: Recognize revenue when or as each obligation is satisfied.** This is where "over time" versus "point in time" decides everything, and it is the subject of the rest of this lesson.

Over Time vs. Point in Time: The Test That Decides Your Revenue Pattern

Point in time means you record revenue when the customer takes control of the deliverable. Deliver 500 laptops, the government accepts them, you book the revenue that day. Simple. Over time means you record revenue progressively while you do the work. ASC 606 allows this when any one of three conditions is true: the customer receives and consumes the benefit as you perform (typical for services), your work creates or enhances an asset the customer controls (typical for work on government-furnished equipment or a government facility), or your work creates an asset with no alternative use to you and you have an enforceable right to payment for work completed to date. That third condition is why most defense development and production contracts book revenue over time. A radar built to a military specification has no alternative buyer, and the termination for convenience clause guarantees the contractor is paid for work performed plus a reasonable profit if the government cancels. No alternative use, enforceable right to payment. Over time it is.

Which Pattern Applies to Your Contract?

CONTRACT SITUATION	TYPICAL PATTERN	WHY
Cost-plus services (CPFF, CPAF, T&M)	Over time	Government consumes the benefit as you perform; you bill for hours and costs as incurred
FFP development of a custom system	Over time	No alternative use, and the T4C clause gives an enforceable right to payment
FFP production of custom hardware	Over time (usually)	Same logic as development; progress payments reinforce the right-to-payment argument
Commercial off-the-shelf items resold to the government	Point in time	Product has alternative use; control transfers at delivery and acceptance
Spare parts from a catalog	Point in time	Standard items, not customized; revenue at shipment or acceptance
Sustainment or maintenance, priced per year	Over time	Government consumes the service continuously; often recognized straight-line or as costs incurred

Cost-to-Cost: Where Your EAC Enters the Income Statement

Once a contract is over time, the company needs a way to measure how far along the work is. The method almost every defense contractor uses is called cost-to-cost. Percent complete equals costs incurred so far divided by total estimated costs at completion. That denominator is your EAC. Revenue to date is then percent complete multiplied by the transaction price. Profit to date is revenue to date minus costs to date. The revenue for this month is simply revenue to date minus everything already recognized in prior months.

IN PLAIN TERMS

Picture a road trip where you judge progress by fuel. You expect the whole trip to take 10 tanks of gas, you have burned 5, so you tell everyone you are halfway there. If the trip really takes 12 tanks, you were never halfway. You only thought so because the total was wrong. Cost-to-cost works the same way: costs spent are the tanks burned, the EAC (Estimate at Completion) is the expected total, and a wrong EAC means the company reports the wrong revenue.

The Cost-to-Cost Method

Percent Complete = Cumulative Costs Incurred ÷ Estimated Costs at Completion (EAC)

Revenue to Date = Percent Complete × Transaction Price

Profit to Date = Revenue to Date – Cumulative Costs Incurred

Every term on the right side of the first line is something the PM controls or forecasts. The finance team does not invent the EAC. They take yours. If your EAC is stale, optimistic, or padded, the company's revenue is wrong by the same amount.

A Worked Example You Can Draw on a Whiteboard

You run a \$20M firm-fixed-price contract to build and deliver a custom ground control system. Your EAC for cost is \$17M, which means expected profit is \$3M, a 15% margin. At the end of the second quarter you have spent \$8.5M. Percent complete is 8.5 divided by 17, which is 50%. Revenue to date is 50% of \$20M, or \$10M. Profit to date is \$10M minus \$8.5M, or \$1.5M. Half the work, half the profit. That is the whole idea. Notice you may have invoiced more or less than \$10M by now. Maybe the contract pays on milestones and you have only billed \$7M. The \$3M difference sits on the balance sheet as an unbilled receivable. It is real revenue, but not yet an invoice and definitely not yet cash. The next lesson covers what happens to that profit when the EAC moves.

The Worked Example at a Glance

\$20M

Contract price
FFP, single performance
obligation

\$17M

EAC (cost)
Your estimate, from the
program

50%

Percent complete
\$8.5M spent ÷ \$17M EAC

\$1.5M

Profit to date
\$10M revenue – \$8.5M cost

THE TRAP: SPENDING FASTER LOOKS LIKE PROGRESS

Under cost-to-cost, every dollar you spend increases percent complete, which increases recognized revenue. If your team burns \$1M on rework that adds nothing, the math still says you are more complete, unless you also raise the EAC by that \$1M. Honest EAC updates are what keep the revenue honest. An EAC that never moves while the burn accelerates is the classic sign of a program quietly building a loss that will surface all at once.

Why This Is the PM's Job, Not Just Finance's

The accounting team can add up costs. They cannot know that a supplier slipped, that the integration is harder than planned, or that the customer is about to ask for a change. Those facts live with you. That is

why most contractors run a monthly or quarterly EAC review where the PM signs off on the estimate. When you sign, you are attesting to the number the company will report. Treat it that way.

One sentence to teach your team: **on an over-time contract, the EAC is the denominator of the company's revenue formula, so the PM who owns the EAC owns the revenue.**

INTERMEDIATE

Cost-Type Contracts: Revenue Goes Up When Costs Go Up, but Fee Does Not

On a cost-plus-fixed-fee contract the transaction price is not fixed. It is estimated cost plus the fixed fee, and it grows if the government funds an overrun. So when the EAC grows by \$1M on a CPFF contract, revenue to date and total expected revenue both grow, but the fee in dollars stays exactly the same. Your margin percentage drops because the same fee is now spread across more cost. Example: \$10M estimated cost, \$700K fixed fee, 7% margin. The EAC grows to \$11M and the government funds it. Revenue at completion is now \$11.7M, fee is still \$700K, margin is now 6%. The company reports more revenue and less profitability. Executives who track margin by business unit notice this immediately. The dangerous version: the EAC grows past the funded ceiling and the government does not add money. Costs above ceiling are still costs, but there is no revenue to match them. On a cost-type contract, over-ceiling work is a pure loss unless you stopped work under the Limitation of Funds clause. The EAC lesson in the Finance module explains the clause; this is why it exists.

How an EAC Increase Hits Each Contract Type

CONTRACT TYPE	REVENUE AT COMPLETION	PROFIT IN DOLLARS	MARGIN %	WHAT THE CFO SEES
FFP	Unchanged	Drops dollar for dollar	Drops sharply	Cumulative catch-up charge this quarter
CPFF (funded)	Rises with cost	Unchanged	Drops	More revenue, weaker margin
CPFF (over ceiling, unfunded)	Capped at ceiling	Drops dollar for dollar above ceiling	Drops sharply	Loss on unfunded work
CPIF	Rises with cost	Drops by contractor share of overrun	Drops	Fee erosion per the share ratio
CPAF	Rises with cost	Base fee unchanged; award fee probably drops	Drops	Lower expected award fee lowers transaction price
T&M	Rises with hours billed	Rises with hours (until ceiling)	Roughly flat	Ceiling is the real constraint

Variable Consideration: Why Award Fee Is Booked Before It Is Awarded

Under ASC 606 the company must estimate award fee and incentive fee and include the portion it expects to earn in the transaction price, as long as a significant reversal is unlikely. That means your award fee is being recognized as revenue every month based on an estimate, typically your historical earn rate. If the Award Fee Board scores you at 70% after finance assumed 85%, the shortfall is reversed in the quarter of the determination. The rating letter you get from the government is also an accounting event. The practical consequence: when you sense a bad award fee period coming, tell finance early. A downward estimate booked in the same quarter you saw it coming is normal. A surprise reversal a quarter later is the kind of thing that ends up in an earnings-call explanation.

Unbilled Receivables and Deferred Revenue: Reading the Balance Sheet Version of Your Program

When recognized revenue runs ahead of billing, the difference is an unbilled receivable, also called a contract asset. When billing runs ahead of recognized revenue, for instance under progress payments or advance milestone billing, the difference is deferred revenue, a contract liability. Neither is a problem by itself. A large and growing unbilled balance is a warning sign though, because it means the company is financing work it has not been paid for. Finance teams watch unbilled aging the way you watch schedule slips. If your program has \$3M unbilled and the next milestone is four months out, expect a question about it.

Questions Finance Will Ask at the Quarterly EAC Review

These are the standard questions. Have the answers before the meeting.

Has the scope changed since the last EAC? SCOPE

Any mod, TDL, or customer request that added work without adding price is an EAC increase you must book now.

Undocumented constructive changes are the most common source of silent EAC growth. If the customer asked for it and you did it, it is either in a mod with price or it is eating your margin.

What is driving the variance between plan and actuals? VARIANCE

Rate variance (people cost more than planned) versus usage variance (more hours than planned) versus timing (work shifted, total unchanged).

Timing variances do not change the EAC. Rate and usage variances usually do. Finance needs you to separate them.

What assumptions are in the estimate to complete? ETC

Productivity going forward, staffing plan, supplier quotes, remaining risk.

If your ETC assumes productivity better than what you have achieved to date, be ready to say why. Finance has seen that assumption fail many times.

Is any of the fee at risk? FEE

Award fee expectations, incentive share, penalty clauses, key personnel deductions.

The answer changes the transaction price, not just the profit. It is a revenue question, not only a profit question.

Do we have a risk of loss? LOSS

On FFP, does the EAC exceed price under any credible scenario?

If yes, the company may need to book a forward loss reserve now. This is the question that gets a program escalated.

ADVANCED

Forward Losses: The One Rule With No Averaging

Everything else in cost-to-cost accounting smooths profit across the life of the contract. Losses do not get that treatment. The moment the EAC on a fixed-price contract exceeds the contract price, ASC 606 requires the company to recognize the entire projected loss immediately, in that period, for the whole contract. Not the portion related to work done so far. All of it. A \$50M FFP program with an EAC of \$53M books a \$3M loss the quarter that EAC is approved, even if the program is only 30% complete. If the EAC later improves to \$52M, the company reverses \$1M. If it worsens to \$55M, another \$2M charge. Public defense primes disclose these individually when they are large enough, which is why you see headlines about a single

fixed-price development program costing a company hundreds of millions in one quarter. That headline is a PM's EAC review, made public.

Performance Obligation Design: Decisions Made at Proposal Time That Fix Your Revenue Pattern for Years

How the contract is structured determines how many performance obligations it has, and that decides whether a problem on one part of the work drags down revenue on the rest. A single-obligation contract covering development, production, and two years of sustainment pools every dollar of cost into one EAC. An overrun in development lowers percent complete for the whole thing, including the profitable sustainment years, and a forward loss on development can wipe out the margin on everything. Separating those into distinct obligations (which usually follows from distinct CLINs with separately identifiable deliverables and prices) means each is measured on its own. The sustainment years keep their margin even if development struggles. Senior PMs and capture leads should involve the company's revenue accounting lead when structuring CLINs on a large proposal. It is one of the few times a contracting decision directly shapes the income statement for the next five years.

Margin Profile: Combined vs. Separated Performance Obligations

STRUCTURE	DEVELOPMENT (EAC OVERRUNS BY \$4M ON \$30M)	SUSTAINMENT (\$20M, 20% MARGIN)	TOTAL REPORTED PROFIT
One combined obligation, \$50M	Loss folded into the pool	Margin diluted by development overrun	About \$1M on \$50M, and a forward loss if the overrun grows
Two separate obligations	Forward loss of \$1M booked immediately	\$4M margin preserved and recognized on its own schedule	\$3M, with the loss isolated and visible

When the Government Changes Its Mind: Contract Modifications Under ASC 606

A modification that adds distinct scope at a price reflecting its standalone value is treated as a new contract. Everything else, including in-scope changes, price adjustments, and REAs, is folded into the existing contract with a cumulative catch-up. An unpriced change order is the difficult case. The company incurs cost immediately but can include the expected price in the transaction price only if it is probable the government will pay and the amount can be reasonably estimated. Until the mod is definitized, the PM's documentation of the change (the constructive change file from the Modifications lesson) is the evidence finance relies on to book any revenue at all. No documentation, no revenue, and the cost still counts. That is the accounting version of "if it is not documented, it did not happen."

WHAT SENIOR PMS DO DIFFERENTLY

They treat the EAC review as a financial control, not a status report. They know which of their contracts are over time versus point in time and why. They know how many performance obligations each contract has. They brief finance on award fee expectations before the board meets. And when a program is heading toward a loss, they raise it in the quarter they see it, because a loss booked early is a management decision and a loss booked late is a restatement.

BUILD ON THIS

- **Estimate at Completion (EAC)** How to build and defend the number that drives everything in this lesson
- **Revenue Recognition** Principal vs. Agent: whether you book the whole contract or just your fee
- **Why Q3 Was Bad** What a cumulative catch-up does to a quarter, with numbers

Key Terms

ASC 606

The U.S. accounting standard (Revenue from Contracts with Customers) that governs when and how much revenue a company may record. Every defense contractor that produces financial statements follows it.

Performance Obligation

A promise in a contract to deliver a distinct good or service. Revenue is recognized as each performance obligation is satisfied. A CLIN is often, but not always, one performance obligation.

Over-Time Recognition

Recording revenue progressively as work is performed, rather than all at once at delivery. Most defense contracts qualify because the work is customized and the government has an enforceable right to pay for work completed.

Point-in-Time Recognition

Recording revenue only when control of the deliverable transfers to the customer, typically at delivery and acceptance. Common for commercial off-the-shelf products sold to the government.

Cost-to-Cost Method

The most common way defense contractors measure progress on over-time contracts: percent complete equals costs incurred to date divided by total estimated costs at completion (the EAC).

Transaction Price

The total amount the contractor expects to be entitled to under the contract, including fee and any variable amounts (award fee, incentives) the company believes it will probably earn.

Variable Consideration

Contract value that depends on future events: award fee, incentive fee, options, penalties. Under ASC 606 it is included in revenue only to the extent it is probable that a significant reversal will not occur.

Unbilled Receivable (Contract Asset)

Revenue the company has recognized but has not yet invoiced. Grows when work is performed faster than billing milestones. It is real revenue on the income statement but not yet cash.

Cumulative Catch-Up

When the EAC changes, the company recalculates total revenue and profit to date and books the entire difference in the current period. A large EAC change lands as a one-quarter hit or gain.

Forward Loss

When the EAC on a fixed-price contract exceeds the contract price, the company must recognize the entire projected loss immediately, not spread it over the remaining work.

Knowledge Check

A \$20M FFP contract has a cost EAC of \$16M. Costs incurred to date are \$4M. Using the cost-to-cost method, how much revenue has been recognized to date?

A. \$4M

✓ B. \$5M

C. \$16M

D. \$20M

WHY

Percent complete is $\$4M \div \$16M = 25\%$. Revenue to date is $25\% \times \$20M = \$5M$. Profit to date is $\$5M - \$4M = \$1M$.

Which condition most often qualifies a custom DoD hardware development contract for over-time revenue recognition?

A. The contract is over \$10M

✓ B. The item has no alternative use and the contractor has an enforceable right to payment for work completed

C. The contractor uses EVM

D. The government pays on delivery

WHY

Custom military hardware has no alternative buyer, and the termination for convenience clause guarantees payment for work performed. Together those satisfy the third over-time criterion in ASC 606.

A contractor resells commercial off-the-shelf laptops to the government with no configuration work. When is revenue typically recognized?

- A. As costs are incurred
- B. When the purchase order is signed

✓ C. At delivery and acceptance (point in time)

- D. Evenly over the contract period

WHY

Standard commercial products have alternative use and control transfers at delivery, so revenue is recognized at a point in time, not over time.

On a CPFF contract, the government funds a \$1M cost overrun. What happens to the contractor's reported numbers?

- A. Revenue and fee both increase

✓ B. Revenue increases, fee in dollars stays the same, margin % falls

- C. Revenue stays the same, fee falls
- D. Nothing changes until the contract closes

WHY

On cost-plus-fixed-fee, revenue is cost plus fee, so funded cost growth raises revenue. The fee is fixed in dollars, so margin percentage drops.

Recognized revenue on your program is \$10M but you have only invoiced \$7M. The \$3M difference is:

- A. Deferred revenue

✓ B. An unbilled receivable (contract asset)

- C. A forward loss
- D. Management reserve

WHY

Revenue recognized ahead of billing creates an unbilled receivable. It is real revenue but not yet an invoice and not yet cash.

Your CPAF contract has a 10% award fee pool. Finance has been recognizing fee at an 85% earn rate. The Award Fee Board scores the period at 65%. What is the accounting result?

A. No change; award fee is only booked when paid

✓ B. The shortfall versus the 85% estimate is reversed in the quarter of the determination

C. The full 10% pool is written off

D. The contract converts to CPFF

WHY

Award fee is variable consideration that is estimated and recognized as work is performed. When the actual determination is lower than the estimate, the difference is reversed in that period.

The EAC on a \$50M FFP contract rises to \$53M when the program is 30% complete. What must the company record?

A. A \$0.9M loss (30% of \$3M)

✓ B. The full \$3M projected loss immediately

C. Nothing until the contract is 100% complete

D. A \$3M reduction in revenue spread over the remaining work

WHY

Forward losses on fixed-price contracts are recognized in full as soon as they are identified, regardless of percent complete.

Under cost-to-cost, your team spends \$1M on rework that adds no scope and you do NOT change the EAC. What happens to recognized revenue?

A. It falls, because rework is unallowable

✓ B. It rises, because percent complete went up

C. It stays the same

D. It is deferred until the rework is accepted

WHY

Every dollar of cost raises percent complete unless the EAC rises too. This is why stale EACs overstate revenue and hide developing losses.

Put the five steps of ASC 606 revenue recognition in order:**CORRECT ORDER**

1. Identify the contract
2. Identify the performance obligations
3. Determine the transaction price
4. Allocate the transaction price to the obligations
5. Recognize revenue when or as each obligation is satisfied

WHY

Identify the contract, identify the performance obligations, determine the transaction price, allocate the price to the obligations, then recognize revenue when or as each obligation is satisfied.

Match each term to its meaning:**CORRECT MATCHES**

Unbilled receivable	Revenue recognized but not yet invoiced
Deferred revenue	Billed to the customer but not yet earned
Cumulative catch-up	One-period adjustment after an EAC change
Variable consideration	Contract value that depends on future events

WHY

Unbilled receivables are recognized but not invoiced. Deferred revenue is billed but not yet earned. Cumulative catch-up is the one-period adjustment from an EAC change. Variable consideration is contract value that depends on future events like award fee.

Why might a capture team deliberately structure development and sustainment as separate CLINs with separate prices?

A. To avoid EVM reporting

✓ **B. So each becomes its own performance obligation and a development overrun does not dilute sustainment margin**

C. Because the FAR requires it

D. To qualify for point-in-time recognition

WHY

Separate performance obligations are measured separately. A forward loss on development is then isolated instead of pooling with, and eroding, the profitable sustainment work.

The government directs extra work verbally; the mod is not yet priced. Your team has incurred \$400K on it. What does finance need from you to recognize any revenue for that work?

A. A signed invoice

✓ B. Documentation showing the change was directed and that payment is probable and estimable

C. DCAA approval

D. Nothing; unpriced work is always booked at cost plus fee

WHY

Unpriced changes can be included in the transaction price only when recovery is probable and the amount can be reasonably estimated. Your constructive change documentation is the evidence. Without it, the cost counts and the revenue does not.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 2

Why Q3 Was Bad: EAC Changes and the Quarterly Results

26 min

How one EAC change on one program becomes a line in the company's quarterly results. The cumulative catch-up math, why the hit is always bigger than the overrun, how public primes report "net EAC adjustments," and what a PM can do to make the quarter boring.

The Meeting Where Everyone Looks at You

It is the second week of October. The business unit is reviewing third-quarter results and the operating margin came in two points under plan. The reason, when the CFO gets to the slide, is one program: yours. The EAC went up \$1.5M in September. On a \$20M contract, that is a 7.5% overrun. Painful, but survivable, you think. Then the slide shows your program's profit for the quarter: negative \$649K, about \$1M worse than plan. And the projected margin on the rest of the program: cut in half. How does a \$1.5M cost increase produce a \$649K loss in one quarter and then keep hurting? That is what this lesson explains. When you understand it, you will understand why finance reacts to EAC changes the way it does, and how to keep your program off that slide.

Quarterly Results in 60 Seconds

Every company closes its books every quarter. Public companies file the results and explain them to investors on an earnings call. Private companies report to their bank, their board, or their owners. Either way, the quarter is the unit of accountability for executives. Annual bonuses, stock prices, loan covenants, and hiring plans all key off quarterly numbers. On the program side, most contractors update EACs monthly and formally lock them once a quarter to feed the close. That quarterly EAC lock is the moment your estimate becomes the company's reported number. Whatever changed in your estimate since the last lock is recognized, in full, in this quarter. That is the cumulative catch-up.

The Math, Step by Step

Pick up the program from the previous lesson: a \$20M FFP contract, cost EAC of \$17M, expected profit \$3M, 15% margin. At the end of Q2 you had spent \$8.5M, you were 50% complete, and you had recognized \$1.5M of profit. During Q3 you spend another \$2M (total \$10.5M), and at the September EAC review you raise the cost EAC to \$18.5M because the integration is taking longer than planned. New expected profit: \$20M minus \$18.5M equals \$1.5M. New margin: 7.5%. New percent complete: \$10.5M divided by \$18.5M equals 56.8%. New profit to date: 56.8% of \$1.5M equals \$851K. Profit already recognized through Q2: \$1.5M. Q3 profit: \$851K minus \$1.5M equals negative \$649K. And that is with \$2M of real work done in the quarter. If you had not changed the EAC, Q3 would have shown a gain of

about \$353K (an extra 11.8% of \$3M). The swing from what was expected to what was booked is roughly \$1M. That is why finance calls it a hit.

Before and After the EAC Change

15% → 7.5% Booking rate Margin on every future dollar is halved	\$1.5M → \$851K Profit to date Recomputed on the new estimate	-\$649K Q3 profit Despite \$2M of work performed	~\$1M Swing vs. plan What the CFO has to explain
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WHY THE HIT IS ALWAYS BIGGER THAN IT FEELS

The catch-up reprices all the work you already did, not just this quarter's. If you were 50% complete at the old margin, half the lost profit was already sitting on the books and has to come back out now. Then the remaining work earns at the new, lower rate. So a mid-program EAC increase hits twice: once as a reversal of past profit, and again as lower profit on everything still to come.

Cumulative Catch-Up

Catch-Up = (New % Complete × New Expected Profit) – Profit Previously Recognized

If the result is negative, the quarter takes a charge. If positive (an EAC improvement), the quarter gets a boost. Both are booked entirely in the period the EAC changes.

The Good Version: Favorable Catch-Ups

The same math works in reverse. Retire a risk, close out a subcontract under budget, or finish testing early, and the EAC drops. The company recomputes profit to date at the higher margin and books the difference as a gain this quarter. This is why mature programs often show rising margins in their last third, and why executives ask PMs to identify "opportunities" alongside risks. An honest EAC reduction is real profit, recognized immediately. A sandbagged EAC that is never lowered is profit the company earned and never got credit for on time.

How This Shows Up in a Public Company's Results

Large defense contractors run hundreds of over-time contracts at once. Every quarter, all of their catch-ups are added together and disclosed as net favorable or unfavorable EAC adjustments (the exact wording varies by company). Analysts watch this number closely because it tells them whether the company's programs are healthy. A quarter where net adjustments are strongly positive suggests a company that is

retiring risk and estimating conservatively. A quarter with a large negative number means something broke, and the company will usually name the program. When you hear that a prime took a \$200M charge on a fixed-price development program, that is one PM's EAC review, multiplied by scale, reported to the market. The mechanics are identical to your \$20M contract.

What a PM Can Do to Make the Quarter Boring

Boring is the goal. Finance loves a program whose EAC moves in small, well-explained steps. Four habits get you there.

Four Habits of PMs Whose Programs Never Make the Slide

- **Update the EAC monthly, not quarterly.** Small monthly moves are absorbed. A quarterly surprise is a catch-up with a story attached. If you see a \$200K problem in July, book it in July.
- **Separate timing from true cost growth.** Work that shifted right but still costs the same is not an EAC change. Work that costs more is. Report both, labeled, so finance can book the right one.
- **Flag risks before they hit the EAC.** A written risk with a dollar range gives finance the option to hold a reserve early. A risk that appears for the first time as an EAC increase looks like the PM was not watching.
- **Report opportunities too.** If a supplier is tracking under budget or a test is likely to pass first time, say so. Favorable catch-ups are the other half of the picture and the part most PMs forget to claim.

One sentence to teach your team: **the quarter does not punish you for an overrun, it punishes you for the surprise, because a surprise reprices every dollar you already earned.**

INTERMEDIATE

Booking Rates and the Conservative Start

Most contractors do not start a new program at the proposed margin. They begin recognizing profit at a lower booking rate, sometimes zero for the first several months of a high-risk development, and raise it as risks retire. This is deliberate. Starting low means most surprises produce small catch-ups, and every risk that closes produces a favorable one. Starting at full margin means every problem is a charge. As a PM, know your program's current booking rate and what events are planned to raise it. Those events are usually tied to your own milestones: a successful PDR, first article acceptance, a supplier delivery. The schedule you manage is also the schedule finance uses to plan margin growth.

A Typical Booking-Rate Path on a Development Program

PHASE	TYPICAL BOOKING RATE	WHAT RAISES IT
Award through first design review	0% to 5%	Requirements baseline agreed, key staff in place
Design complete through first hardware	5% to 8%	PDR and CDR passed, long-lead parts on contract
Integration and test	8% to 11%	First article accepted, major test events passed
Final delivery and closeout	Full proposed margin, often higher	Remaining risk reserve released; subcontract closeouts under budget

Rate Variance vs. Volume Variance: Finance Needs You to Split Them

When actuals exceed plan, the cause is one of three things: you used more hours or material than planned (volume), each hour or part cost more than planned (rate), or work happened earlier than planned (timing). Only the first two change the EAC. Timing changes the shape of the spend curve but not the total. A common mistake is reporting "we are \$300K over plan" without the split. Finance then has to guess how much of that is a real EAC change, and they will guess conservatively, which means a bigger charge than necessary. Come to the review with three numbers, not one.

Private Companies: No Stock Price, Same Pressure

If you work for a private contractor, there is no earnings call, but the quarter still matters. Bank loans carry covenants, often a minimum trailing profit or a maximum ratio of debt to earnings. A large unfavorable catch-up can breach a covenant even when the company has plenty of cash, triggering a renegotiation with the bank at bad terms. Private-equity-owned contractors report to a board that tracks EBITDA against an investment case. And for any company preparing to be sold, the trailing twelve months of profit is what the price is based on. A \$650K charge on your program can move the sale price by several times that amount, because buyers pay a multiple of earnings.

ADVANCED

Management EAC vs. Accounting EAC: Which Number Is Being Asked For

Many large contractors keep two estimates. The management EAC is the PM's most likely outcome, including planned risk mitigation and expected opportunities. The accounting EAC, used for revenue recognition, is typically more conservative: it includes identified risks at their expected value and excludes opportunities until they are highly probable. The difference is often held as a program-level reserve inside the EAC. Senior PMs need to speak both. In a program review, the management EAC drives decisions. In

the quarterly close, the accounting EAC drives the numbers. Presenting the optimistic one to finance produces charges later. Presenting the conservative one to the customer invites questions about why you need so much money. Know which room you are in.

Portfolio Effects: Why One Program Can Move a Segment

Segment margins in defense typically run in the 9% to 12% range. On a \$2B segment, one point of margin is \$20M. A single fixed-price development program with a \$20M unfavorable catch-up therefore moves the whole segment by a point, and if the company guided to a range, it may have to lower guidance. That is the chain from a PM's EAC to a headline: EAC review, catch-up, segment margin, guidance, stock price. Senior PMs at large primes should know their program's size relative to the segment, because it tells them how much scrutiny an EAC change will get. A \$1M swing on a \$50M program inside a \$2B segment is a footnote. The same \$1M swing inside a \$150M business unit at a mid-size contractor is a board conversation.

Reading a Prime's Quarterly Disclosure on EAC Adjustments

Public defense contractors describe program profit adjustments in their quarterly filings. Here is how to read that section like an analyst.

Net EAC adjustments (total) HEADLINE

Favorable minus unfavorable across all contracts. Compare to the same quarter last year and to segment operating profit.

If net adjustments are a large share of segment profit, the company's earnings depend on risk retirement rather than base performance. That is normal late in a program cycle and worrying early in one.

Gross favorable and gross unfavorable DETAIL

The two components separately. A small net number can hide a large charge offset by many small gains.

Rising gross unfavorable with flat net means the company is using favorable adjustments elsewhere to cover problem programs.

Named program charges PROGRAMS

Charges large enough to be identified by name, usually fixed-price development.

Look for the phrase "reach-forward loss" or "forward loss." That is the whole projected loss on a contract booked at once, per the previous lesson.

Backlog change CONTEXT

A charge on a program that also shrinks backlog (descoping) is a different story from a charge on a program the company is still growing.

Backlog is covered in its own lesson in this module. Read the two sections together.

WHAT SENIOR PMS DO DIFFERENTLY

They know the company's quarterly close calendar and get their EAC to finance before the deadline, with the rate, volume, and timing split already done. They keep a written risk register with dollar ranges so finance can reserve early. They present a management EAC and an accounting EAC and explain the gap. And when a charge is unavoidable, they book it in the quarter they see it, with a recovery plan attached, because the market forgives a charge and punishes a pattern of surprises.

BUILD ON THIS

- **Revenue Recognition II** The cost-to-cost method this lesson builds on
- **Risk Management** The risk register that lets finance reserve before the EAC moves
- **The R&O** Every risk and opportunity in dollars, rolled up so nothing is a surprise
- **What Your CFO Sees** Gross margin, operating margin, and the language of the business review

Key Terms

Quarter (Q1 to Q4)

A three-month financial reporting period. Public companies report results every quarter; private companies usually report to their bank and board on the same cadence. Program EACs are typically locked once per quarter to feed these results.

Cumulative Catch-Up Adjustment

The one-time change to revenue and profit recorded when an EAC changes. It brings profit-to-date in line with the new estimate all at once, so it includes the effect on all past work, not just this period's.

Net EAC Adjustments

The total of favorable and unfavorable catch-ups across all programs in a period. Public defense contractors disclose this number; it is one of the first things analysts look at.

Profit Booking Rate

The margin percentage a company is currently recognizing on a contract, equal to expected profit divided by expected revenue at completion. It changes every time the EAC changes.

Risk Retirement

Raising the booking rate on a program as risks are resolved (a successful test, a supplier delivering). This produces favorable catch-ups later in the contract and is why margins often rise as a program matures.

Management EAC vs. Accounting EAC

Some companies keep a risk-adjusted internal EAC for management and a separate, more conservative EAC for revenue recognition. The PM needs to know which one is being asked for.

Earnings Call

The quarterly call where a public company's executives explain results to investors and analysts. Large program charges are named individually on these calls.

Segment Margin

Operating profit as a percentage of revenue for one business segment (for example, Aeronautics or Mission Systems). A single program charge can move a segment's margin by a full point.

Guidance

The revenue and profit range a public company tells investors it expects for the year. A large unfavorable EAC adjustment can force the company to lower guidance, which is what makes the stock fall.

Covenant

A financial condition in a loan agreement, such as a minimum profit level or maximum debt ratio. For private contractors, a bad quarter can breach a covenant even when the company is fundamentally healthy.

Knowledge Check

A \$20M FFP contract had a cost EAC of \$17M and \$1.5M of profit recognized through Q2 at 50% complete. In Q3 costs reach \$10.5M and the EAC rises to \$18.5M. What is profit to date after the change?

A. \$1.5M

✓ B. \$851K

C. \$649K

D. \$2.1M

WHY

New expected profit is \$1.5M. Percent complete is $\$10.5M \div \$18.5M = 56.8\%$. Profit to date is $56.8\% \times \$1.5M =$ about \$851K.

In the same example, what is the profit recognized in Q3 alone?

A. +\$351K

✓ B. -\$649K

C. -\$1.5M

D. \$0

WHY

Profit to date after the change is \$851K. Profit already recognized was \$1.5M. Q3 profit is $\$851K - \$1.5M = -\$649K$, even though \$2M of work was done.

Why is a mid-program EAC increase said to "hit twice"?

A. Because DCAA audits it twice

✓ B. Because past profit is reversed and future work earns at a lower rate

C. Because it is booked in two quarters

D. Because both revenue and cost are restated

WHY

The catch-up reprices work already performed (a reversal now) and the remaining work is recognized at the new lower booking rate (less profit later).

An EAC decreases because a subcontract closed out \$400K under budget. What happens this quarter?

A. Nothing until the contract ends

✓ B. A favorable cumulative catch-up is recognized immediately

C. The \$400K is returned to the government

D. The booking rate is frozen

WHY

Favorable EAC changes use the same math as unfavorable ones. Profit to date is recomputed at the higher margin and the difference is booked as a gain in the current period.

Which of these changes the EAC?

A. Work that shifted from Q3 to Q4 at the same cost

✓ B. Hours that came in 10% above plan for the same scope

C. A milestone billing that was submitted late

D. A change in the quarterly close date

WHY

Only true cost growth (volume or rate variance) changes the EAC. Timing shifts change the spend curve, not the total.

Why do many contractors start a new development program at a booking rate below the proposed margin?

A. The FAR requires it

✓ B. So early surprises produce small charges and retired risks produce favorable catch-ups later

C. Because DCAA disallows early profit

D. To reduce the transaction price

WHY

A conservative starting rate limits the size of early unfavorable adjustments and creates favorable adjustments as risk is retired, which is the pattern investors prefer.

A public prime discloses large net favorable EAC adjustments that make up most of a segment's profit. What does an analyst likely conclude?

A. The segment is losing money

✓ **B. The segment's earnings depend heavily on risk retirement rather than base performance**

C. The company has no fixed-price contracts

D. Backlog is growing

WHY

When net adjustments are a large share of profit, earnings quality depends on continuing to retire risk. That is normal late in a program cycle and a concern early in one.

You work for a private contractor. Why can a large unfavorable catch-up still be a serious problem even if the company has plenty of cash?

A. It cannot; only public companies care

✓ **B. It can breach a loan covenant tied to profit and force a renegotiation with the bank**

C. It triggers a DCAA audit

D. It voids the contract

WHY

Bank covenants often key off trailing profit or debt-to-earnings ratios. A big charge can breach a covenant regardless of cash on hand.

Order the chain from a PM's estimate to a company headline:

CORRECT ORDER

1. PM raises the program EAC at the quarterly review
2. Finance books a cumulative catch-up charge
3. Segment operating margin drops
4. Company lowers annual guidance
5. Stock price falls and the program is named on the earnings call

WHY

The PM raises the EAC, the quarterly close books a cumulative catch-up, that moves the segment margin, the company lowers guidance, and the stock reacts.

Match the variance type to its effect on the EAC:**CORRECT MATCHES**

Volume variance	More hours or parts than planned; raises the EAC
Rate variance	Each hour or part costs more; raises the EAC
Timing variance	Work moved earlier or later; EAC unchanged

WHY

Volume and rate variances are real cost growth and change the EAC. Timing variance shifts when cost occurs but not how much.

What is the difference between a management EAC and an accounting EAC?

A. There is no difference

✓ **B. The management EAC includes expected opportunities; the accounting EAC is more conservative and drives revenue recognition**

C. The accounting EAC is set by the government

D. The management EAC excludes labor

WHY

Many contractors keep a most-likely internal estimate for decisions and a conservative estimate for the books. Senior PMs must know which is being requested.

You identify in July that integration will likely cost \$200K more than planned. Which action best protects the company's quarter?

A. Wait for the September lock to see if it resolves

✓ **B. Raise the EAC or document the risk with a dollar range in July so finance can reserve early**

C. Absorb it in management reserve without reporting

D. Report it to the government first

WHY

Early, documented risk lets finance reserve or book a small adjustment now. A surprise at the quarterly lock becomes a larger catch-up with a story attached.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 3

The R&O: Every Risk and Opportunity to Revenue, on One Page

24 min

The industry-standard tool that keeps nobody in the dark. How contractors put a likelihood and a dollar value on every risk and opportunity, roll them up so a leader sees the whole board, and decide when one program's good news really cancels another's bad news.

Nobody Likes Being the Last to Know

Two program managers (PMs) walk into the monthly business review. One has a \$600K problem. The other has a \$400K opportunity she has not mentioned yet, because it is not certain and a little cushion feels nice. The business unit leader hears only the problem. She calls the chief financial officer (CFO) and warns about a soft quarter. Six weeks later the opportunity lands and the quarter is fine. Nobody made a bad decision with bad data. They made one with half the data. The fix is a tool most defense contractors already run: the R&O, short for Risks and Opportunities. It puts every risk and every opportunity on one page, in dollars, with odds attached, where everyone who needs to see it can.

IN PLAIN TERMS

Think of a household planning next month's budget. One partner expects the car will probably need a big repair. The other expects a possible bonus but has not mentioned it because it is not certain. Planning around only the repair makes the month look worse than it likely is. Writing both on one list, each with its rough odds and dollar amount, shows the real picture. The R&O (Risks and Opportunities) does that for a business unit.

RISK REGISTER VS. R&O: SAME IDEA, DIFFERENT CURRENCY

The Risk Management lesson taught the program risk register: probability and impact scored 1 to 5, a heat map, mitigation owners. That register asks "how bad is this for the program?" The R&O asks "what does this do to the money, and when?" Same items, often the same owners, but scored in percentages and dollars so they can be added up, and with opportunities carried right beside the risks instead of in a footnote.

The Columns That Matter

Formats vary by company, but a working R&O line almost always has these.

- **ID and type.** Risk or opportunity, plus a short ID. Many shops prefix the program so the roll-up stays readable (A-R1, B-O2).
- **What would happen.** Specific enough that someone else could tell whether it happened. "Supplier issues" is not an item. "Sole-source supplier raises RF module prices at the Q4 renewal" is.
- **Likelihood.** A percentage. If your company uses bands (low, medium, high), know the percentage behind each band, because the math needs a number.
- **Impact.** Dollars of profit, and often revenue too, if it happens. Not a 1 to 5 score.
- **Weighted value.** Likelihood times impact. This is what gets rolled up.
- **When.** The quarter it would hit. An opportunity next year does nothing for this quarter.
- **Owner and handling plan.** A named person and what they are doing about it, with a trigger date. No owner, no plan, no line.
- **In the EAC?** Whether the item is already reflected in the booked estimate at completion (EAC). This stops the same dollar from being counted twice.

Weighted Value

Weighted Value = Likelihood × Impact

Add the weighted values of all opportunities, subtract the weighted values of all risks, and you have the net R&O: the forecast's best guess of how the open items will land.

Sample R&O: One Business Unit, September Review

ID	WHAT COULD HAPPEN	LIKELIHOOD	IMPACT (\$K PROFIT)	WEIGHTED (\$K)	WHEN	OWNER
A-R1	Risk: integration testing on Program A runs six weeks long (extra labor)	60%	−600	−360	Q3	Program A PM
A-R2	Risk: sole-source supplier raises RF module prices	30%	−400	−120	Q4	Supply chain lead
B-R1	Risk: Program B's option year starts two months late under a continuing resolution (fee lost this year)	50%	−150	−75	Q4	Program B PM
C-R1	Risk: two senior engineers leave Program C; replacements ramp slowly on the fixed-price CLIN	25%	−200	−50	Q4	Program C PM
A-O1	Opportunity: Program A's cabling subcontract closes out under budget	75%	+400	+300	Q3	Subcontracts manager
B-O1	Opportunity: Program B's award fee period is rated Excellent instead of the planned Very Good	40%	+250	+100	Q4	Program B PM
C-O1	Opportunity: automated ticketing cuts help desk labor on Program C's fixed-price CLIN	50%	+160	+80	Q4	Program C PM

The Board at a Glance

−\$1.35M Gross risk If every risk hits in full	−\$605K Weighted risk Sum of likelihood × impact	+\$480K Weighted opportunity Sum of likelihood × impact	−\$125K Net R&O The forecast's best guess
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What the Leader Sees That No Single PM Can

Look at A-R1 on its own and Program A has a \$600K scare. Look at the whole board and the same program has a cabling opportunity worth +\$300K weighted in the same quarter. Q3 nets to about −\$60K. That is a very different phone call to the CFO. Q4 has five smaller items: three risks worth −\$245K weighted and two opportunities worth +\$180K. Net about −\$65K. Across both quarters the business unit is looking at

roughly –\$125K of net exposure, not the \$600K headline. That is the point of the R&O. Each PM sees one program. The leader sees whether someone else's good news already covers your bad news, where to put effort (make sure A-O1 closes in Q3), and what could still go badly wrong (–\$1.35M if everything breaks the wrong way).

WHEN AN OPPORTUNITY REALLY CANCELS A RISK

Netting is powerful and easy to abuse. An opportunity only offsets a risk in the forecast when four things line up. Same period: a Q1 opportunity does nothing for a Q3 miss. Comparable odds: a 90% risk against a 20% opportunity can have matching weighted values and still leave you exposed most of the time. A real owner and capture plan: an opportunity nobody is working is a wish. And the gross stays visible: net tells you the likely landing, gross tells you how hard the landing could be.

Four Habits of an R&O People Trust

- **Update it with the EAC, every month.** The R&O and the EAC review are the same conversation. Items that move the estimate should show up on the R&O first, while there is still time to act.
- **Every line has a name and a date.** An owner, a handling plan, and a trigger date. Items without them get cut or fixed at the next review.
- **Retire items when the trigger passes.** A risk whose date came and went without happening should leave the list, and any reserve held for it should be released. A stale R&O is worse than none, because people trust it.
- **Give opportunities the same rigor as risks.** No sandbagging and no wishful thinking. If it is 75% likely, it goes on the page at 75%, where the leader can see it and help land it.

One sentence to teach your team: **an R&O turns "I'm worried about" into a dollar figure with odds attached, so the person above you can see whether someone else's good news already covers your bad news.**

INTERMEDIATE

In the EAC or Outside It?

Companies differ on how R&O items relate to the booked EAC, so ask your finance lead which policy yours follows. A common pattern: identified risks are carried in the EAC at their weighted value, while opportunities stay outside it until they are highly likely, because booking profit on hope is how favorable catch-ups turn into unfavorable ones. That is the gap between the management EAC and the accounting EAC from the Why Q3 Was Bad lesson. Whatever the policy, the "In the EAC?" column matters. Without it, a risk gets counted once in the estimate and again on the R&O, and the leader sees a problem twice as big as it is.

Program A's EAC Range, Built From Its R&O

CASE	HOW IT IS BUILT	COST EAC	PROFIT ON A \$20M PRICE	MARGIN
Worst	Baseline \$17.0M plus every risk at full value (A-R1, A-R2)	\$18.0M	\$2.0M	10.0%
Most likely	Baseline plus weighted risks, minus weighted opportunity (+\$360K +\$120K –\$300K)	\$17.18M	\$2.82M	14.1%
Best	Baseline minus every opportunity at full value (A-O1)	\$16.6M	\$3.4M	17.0%

Running the Monthly R&O Review

A good R&O review takes fifteen minutes per program and ends in decisions, not a readout. The leader walks the board by quarter and asks the same questions every month. What is new since last month? What moved, and what data moved it? Which items trigger before the next review? Which opportunities need help from outside the program to land? Is anything sitting at the same likelihood for three months with no change in the handling plan? That last question catches the two classic failures: a risk nobody is working, and an opportunity someone is quietly holding back.

ADVANCED

Is the Mitigation Worth It?

The R&O turns mitigation into a money decision. Take A-R2: a 30% chance of a \$400K supplier price increase, weighted at \$120K. Qualifying a second source costs \$60K and cuts the likelihood to 10%. Before: \$120K expected cost. After: 10% of \$400K is \$40K, plus the \$60K you spent, equals \$100K. Worth it, barely, on expected value. It also cuts the chance of a \$400K hit from three in ten to one in ten, which is often the stronger argument when the program has little margin to spare. The same math works on opportunities: spending \$20K on an early closeout audit that lifts A-O1 from 75% to 90% moves its weighted value from +\$300K to +\$360K, a \$40K gain after the cost.

WHAT PORTFOLIO NETTING CAN AND CANNOT DO

Netting happens in the forecast, not in the books. Revenue and profit are recognized contract by contract, so if C-O1 lands and A-R1 hits, Program A still books its own unfavorable catch-up and Program C books its own favorable one. They cancel only in the business unit total. That is still worth a great deal: the quarter lands where the leader said it would, and nobody has to explain a miss. But a PM cannot point at another program's opportunity and skip reporting a charge on their own.

Calibration: Do Your 75% Items Happen 75% of the Time?

A mature R&O keeps score. At year end, pull every closed item and group them by the likelihood they carried. If the items rated 75% came true about three times in four, the R&O is calibrated and leaders can trust the weighted numbers. If your 75% opportunities landed a third of the time, the business unit has been forecasting with optimism, and every net R&O it reported was too rosy. Most organizations are overconfident on opportunities and underconfident on the risks they have lived with for a while. Senior leaders check this, and PMs whose numbers hold up get believed faster in the next review.

WHAT SENIOR LEADERS DO DIFFERENTLY

They walk the board by quarter, not by program. They ask for gross and net every time. They challenge likelihoods with data ("what happened last time we assumed that?"). They move resources toward the opportunities that can cover the biggest risks, and they reward PMs who report early, because a risk on the R&O in July is a managed item, and the same risk discovered in September is a catch-up with a story attached.

BUILD ON THIS

- **Why Q3 Was Bad** The catch-up an unreported risk turns into
- **Risk Management** The program risk register and the RIO process behind the R&O
- **Estimate at Completion (EAC)** The estimate every R&O item feeds

Key Terms

R&O (Risks and Opportunities)

The contractor's running list of everything that could move a program's cost, revenue, or profit, good or bad, each with a likelihood, a dollar impact, an owner, and a date. Most defense contractors keep one per program and roll them up to the business unit.

Likelihood

The chance, as a percentage, that a risk or opportunity actually happens. Written as a number (60%), not a feeling ("medium"), so it can be multiplied.

Impact

What the item does to the money if it happens, in dollars. Most R&Os state it as profit impact; many also track revenue impact, because a delay can move revenue without changing total profit.

Weighted Value

Likelihood times impact. Also called factored or expected value. A 60% chance of a \$600K hit has a weighted value of \$360K. This is the number that gets added up.

Gross vs. Net

Gross is the total if every risk hits (or every opportunity lands) at full value. Net is weighted opportunities minus weighted risks. Leaders need both: net for the forecast, gross for the worst case.

Handling Plan

What the owner is doing about the item. For a risk, the mitigation that lowers likelihood or impact. For an opportunity, the capture plan that makes it more likely to land.

Trigger Date

The date or event when you will know whether the item happened. After it passes, the item is retired or becomes real (an issue, or a realized gain).

Roll-Up

Adding every program's R&O into one business unit or company view, so a leader can see the whole board by quarter instead of one program at a time.

EAC Range

Worst case, most likely, and best case estimates at completion, built from the R&O. Worst adds every risk at full value, best subtracts every opportunity at full value, most likely uses the weighted values.

Sandbagging

Keeping a likely opportunity off the R&O as a private cushion. It hides profit from the people making decisions and delays when the company gets credit for it.

Knowledge Check

What does an R&O show that a typical 1-to-5 program risk register usually does not?

- A. A list of program risks
- ✓ B. Risks and opportunities side by side, in dollars and percentages, so they can be added up and rolled up to leaders
- C. Only risks rated high
- D. The contract's CLIN structure

WHY

The R&O scores the same kinds of items in percentages and dollars and carries opportunities beside risks, so a leader can total them by quarter across programs.

A risk has a 40% likelihood and a \$500K profit impact. What is its weighted value?

- A. \$500K
- ✓ B. \$200K
- C. \$300K
- D. \$40K

WHY

Weighted value is likelihood times impact: $40\% \times \$500K = \$200K$.

In the sample R&O, A-R1 is **-\$360K weighted** and A-O1 is **+\$300K weighted**, both in Q3. What is Program A's net for Q3?

- A. -\$660K
- ✓ B. -\$60K
- C. +\$60K
- D. -\$360K

WHY

+\$300K minus \$360K is -\$60K. The \$600K headline risk nets to a much smaller Q3 exposure once the same-quarter opportunity is on the page.

The net R&O is only –\$125K. Why should the leader still look at the gross risk of –\$1.35M?

A. Gross is what gets booked

✓ B. Net is the likely landing; gross shows how bad it gets if the risks hit and the opportunities do not

C. Gross is required by the FAR

D. It should not; net is enough

WHY

Weighted values describe the expected outcome. The gross number is the downside the business has to be able to absorb.

A Program B opportunity is expected to land in Q1 of next year. Can it offset a Program A risk that hits this Q3?

A. Yes, any opportunity offsets any risk

✓ B. Not for this quarter: an offset only works in the same reporting period

C. Yes, if both are over 50% likely

D. Only if both programs are fixed-price

WHY

Timing has to match. A later opportunity helps the year or the next quarter, not the quarter the risk lands in.

A 90% likely risk and a 20% likely opportunity have the same weighted value. What is the problem with calling them a wash?

A. There is no problem

✓ B. Most of the time the risk hits and the opportunity does not, so the business is exposed far more often than the net suggests

C. Weighted values cannot be compared

D. Opportunities cannot be weighted

WHY

Equal weighted values hide very different odds. Good netting pairs items with comparable likelihood and timing.

If Program C's opportunity lands and Program A's risk hits, how does the accounting treat them?

A. The opportunity is moved to Program A to cancel the charge

✓ B. Each contract books its own catch-up; they cancel only in the business unit total

C. Neither is booked until year end

D. The risk is ignored because the net is positive

WHY

Revenue and profit are recognized contract by contract. Portfolio netting makes the total land as forecast, but each program still reports its own result.

A second source costs \$60K and cuts a \$400K risk from 30% to 10% likely. On expected value, is it worth doing?

A. No, it costs money

✓ B. Yes: expected cost falls from \$120K to \$100K (\$40K weighted plus the \$60K spent)

C. Only if the risk is above 50%

D. It makes no difference

WHY

Before, $30\% \times \$400K = \$120K$. After, $10\% \times \$400K = \$40K$, plus \$60K spent = \$100K. It also cuts the chance of the full hit from 3 in 10 to 1 in 10.

Put one R&O item's life in order:

CORRECT ORDER

1. PM identifies a specific risk or opportunity
2. Likelihood and dollar impact are estimated
3. An owner, handling plan, and trigger date are assigned
4. The item rolls up to the business unit view
5. It is updated monthly at the EAC review
6. It is retired or realized when the trigger date passes

WHY

An item is identified, sized, given an owner and plan, rolled up for leaders, updated each month with the EAC, and retired or realized when its trigger passes.

Match the R&O term to what it means:**CORRECT MATCHES**

Weighted value	Likelihood times impact
Net R&O	Weighted opportunities minus weighted risks
Gross risk	Every risk at full value
Trigger date	When you will know if the item happened

WHY

Weighted value is likelihood times impact, net is weighted opportunities minus weighted risks, gross is the full-value total, and the trigger date is when you will know.

Program A has a \$20M price and a \$17.0M baseline cost EAC. Its risks are \$600K and \$400K at full value. What is worst-case profit?

A. \$3.0M

✓ B. \$2.0M

C. \$2.82M

D. \$1.0M

WHY

Worst case adds every risk at full value: $\$17.0M + \$0.6M + \$0.4M = \$18.0M$. Profit is $\$20M$ minus $\$18.0M = \$2.0M$, a 10% margin.

A PM keeps a 90% likely \$300K opportunity off the R&O as a private cushion. What is the main harm?

A. None, it is conservative

✓ B. Leaders decide with half the picture, may escalate problems that are already covered, and the company gets credit for the profit late

C. It violates the FAR

D. It lowers the booking rate permanently

WHY

Sandbagging feels safe to the PM but hides information from the people deciding where to spend effort and what to tell the CFO.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 4

Pass-Through vs. Value-Add: The Same Question, Asked by Your Auditor and Your CO

27 min

What separates an integrator that adds value from one that just marks up other people's work. The government's own rule on excessive pass-through charges, the accounting rule on Principal vs. Agent, why they ask the same question, and what it costs you when the answer is "not much."

Two Companies, Same Contract, Very Different Businesses

Company A wins a \$30M contract to deliver and support a secure network for a base. It hires a subcontractor to install the hardware, another to run the help desk, and a third to do the cybersecurity work. Company A's own staff: a PM, a contracts person, and someone who forwards invoices. Company B wins the identical contract. It also uses subcontractors for installation and help desk, but its own engineers design the architecture, write the security plan, integrate the pieces, run acceptance testing, and are on the hook if the system fails. On the surface both are "the prime." Both sign the contract, both bill the government \$30M. But two very different people are going to ask both companies the same question, and the answer determines what the contract is worth to each of them. The government's contracting officer asks: what value do you add for the markup you charge on your subcontractors? The company's auditor asks: do you control this service, or are you just arranging it? Company B has a good answer. Company A does not, and it will cost them in two places at once.

IN PLAIN TERMS

Think of two home renovation contractors. The first draws the plans, schedules the plumber and electrician, inspects their work, and comes back if the kitchen leaks. The second just passes along the plumber's and electrician's phone numbers and adds a fee to their bills. Both technically "ran the job," but only the first earned the markup. The government and the auditor ask the same thing about a prime: did you manage the subcontracted work, or just pass the bill through?

ONE IDEA, THREE RULEBOOKS

There is nothing wrong with subcontracting. Every large program does it. The problem is a prime whose only real function is holding the contract. The government does not want to pay a markup for that, the accounting rules will not let the company book revenue for it, and a buyer will not pay for it when the company is sold. Three different rulebooks, one underlying idea: you get paid for the work you actually do.

The Government's Rule: Limitations on Pass-Through Charges

Congress wrote this rule into law after finding that some primes were adding a layer of cost to work they barely touched. The rule lives in the FAR as the "limitations on pass-through charges" clauses (FAR 52.215-22 and 52.215-23, with a DoD companion clause at DFARS 252.215-7004). Because it is statutory, it survives the current FAR rewrite; verify the exact clause numbers and dollar thresholds in the current text before you cite them. Here is how it works. When you propose on a covered contract and you plan to subcontract more than 70% of the total cost of the work, you must say so in the proposal, identify the amount and percentage, and describe the value you add to that subcontracted work. The contracting officer decides whether your added value is real. If the CO determines you add no or negligible value, the indirect costs and profit you apply to the subcontracted portion become "excessive pass-through charges" and are unallowable. You still pay your subs. You just do not get paid the markup on them. The rule also reaches down: if your subcontractor in turn subcontracts more than 70% of its work, the same disclosure and test apply to it.

What Counts as Added Value (in the Government's Words)

The FAR defines added value as subcontract management functions the CO determines benefit the government. Its own examples are the ones below. Notice they are activities, not titles.

- **Processing orders of parts or services** Doing the actual procurement work: sourcing, ordering, expediting.
- **Maintaining inventory** Holding and managing stock so the government does not have to.
- **Reducing delivery lead times** Getting things to the customer faster than they could get them directly.
- **Managing multiple sources for contract requirements** Coordinating several suppliers into one coherent delivery.
- **Coordinating deliveries** Sequencing, scheduling, and de-conflicting what arrives when.
- **Performing quality assurance functions** Inspecting, testing, accepting, and rejecting subcontractor work on the government's behalf.
- **Technical direction and integration (in practice)** Not listed word for word in the definition, but the strongest case you can make: your engineers design the solution and make the subcontracted pieces work together.

The Accounting Rule: Principal or Agent

The Revenue Recognition lesson in the Finance module introduced this. The auditor's question is whether your company controls the good or service before it transfers to the government. If yes, you are a Principal and you book the full \$30M as revenue. If you merely arrange for someone else to provide it, you are an Agent and you book only your fee, perhaps \$2M. The indicators auditors use line up almost exactly with the government's added-value list. Are you primarily responsible for fulfilling the promise? Do you carry inventory or performance risk? Do you set the price? Do you direct the subcontractor's work and accept or

reject it? Company B answers yes to all four. Company A answers yes only to "we signed the contract," which is not enough.

Same Facts, Two Rulebooks

WHAT YOU ACTUALLY DO	CO'S VIEW (FAR PASS-THROUGH RULE)	AUDITOR'S VIEW (ASC 606)	RESULT
Sign the contract, forward invoices, add markup	No or negligible added value; markup on subs unallowable	Agent; book only your fee	Lose the markup AND report net revenue
Manage subs administratively, some QA	Borderline; depends on documentation and CO judgment	Often Agent; control not clearly demonstrated	Risky on both fronts
Design, integrate, direct, test, accept, warrant	Added value clearly demonstrated; markup allowable	Principal; book gross revenue	Paid for the work and credited for it

Company A's Bill for Being a Pass-Through

Put numbers on it. Company A subcontracts \$25M of the \$30M. It proposed 12% G&A and 8% profit on that subcontracted work, about \$5M of markup. If the CO finds negligible added value, up to that \$5M becomes unallowable. Company A still owes its subs \$25M. Its revenue for accounting purposes, if the auditor concludes it is an Agent, is not \$30M but whatever fee it clears, maybe \$2M to \$3M. The company that thought it had a \$30M contract has a \$3M business with \$25M of other people's money moving through it. Company B subcontracts \$18M and self-performs \$12M with its own engineers. Its markup on the subs is allowable because it demonstrably manages and integrates them, its self-performed work carries full margin, and it books the entire \$30M as revenue. Same contract. One company built something. The other rented its signature.

Company A vs. Company B on a \$30M Contract

\$25M / \$18M Subcontracted A passes through 83%; B 60%	Up to \$5M A's markup at risk Unallowable if no added value	\$3M vs \$30M Revenue booked A as Agent vs B as Principal	\$12M B self-performs Full margin, clear control
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THE HARDWARE DUMP, REVISITED

The classic pass-through is a SOW that says "deliver 50 servers" and a prime that orders them from a distributor. The fix is not clever wording. If the government actually needs the servers configured, hardened, integrated, and tested, then the SOW should say so and your people should do it. Language in a proposal that describes value you do not deliver fails the CO test, fails the audit, and can become a false claims problem. Do the work, then describe the work.

One sentence to teach your team: **markup is payment for management, and if nobody at your company is managing the subcontracted work, the government will not pay for it and the auditor will not let you count it.**

INTERMEDIATE**How the Rule Shows Up in a Proposal and in an Audit**

In the proposal, the pass-through clause requires a specific disclosure when you cross the 70% line: the total cost of work, the amount and percentage you intend to subcontract, and a description of the added value you provide. Evaluators and pricing analysts read that description. A vague paragraph about "program management oversight" invites a determination of negligible value. A specific description that names the functions (integration lab, acceptance test plan, supplier quality audits, configuration management) and the people who perform them is what passes. After award, the question comes back in two places. DCAA can question the indirect costs and profit on subcontracted work during an incurred cost audit if the added value was never demonstrated. And the CO can, at any point, determine that charges are excessive and disallow them. Keep the evidence: subcontract technical direction letters, QA records, integration test results, and staffing records showing who did what.

Subcontract Handling Rates: Recovering the Real Cost of Managing Subs

Many contractors apply their full G&A rate to subcontract costs, which can be 10% or more on large sub dollars. That is often more than the actual cost of managing those subs, and it is exactly the markup the pass-through rule targets. A common response is a separate, lower subcontract handling (or material handling) rate that recovers only the cost of the procurement, subcontract administration, and quality functions that touch subs. A 3% to 5% handling rate on subs, with G&A applied to the smaller self-performed base, prices the work more competitively and is easier to defend as added value because the rate itself is built from the cost of the added-value functions. This is a company-level cost accounting decision, not one a PM makes alone. But a PM should know which structure the company uses, because it changes how competitive the price is on sub-heavy work and how strong the added-value story is.

Margin Reality: Self-Performed vs. Subcontracted Dollars

TYPE OF WORK	TYPICAL PROFIT ON THE DOLLAR	WHAT SUPPORTS THE MARGIN	RISK CARRIED
Self-performed labor	8% to 15%	Your people, your expertise, your performance risk	Full
Managed subcontract (real integration and direction)	5% to 8%	Documented management, QA, integration	Shared; prime still liable to the government
Administrative pass-through	0% to 3% at best; may be disallowed	Little beyond holding the contract	Prime liable, sub in control

Workshare: The Number Everyone Fights About

Because self-performed dollars carry more margin and support the Principal and added-value arguments, primes want to keep as much work as they can and give teammates the rest. Teammates, for the same reasons, want a larger share. The teaming agreement sets a workshare percentage, but the prime usually controls the final split at proposal time, and "workshare commitments" in teaming agreements are often best-effort rather than binding. Understand that when a prime offers you 49% of the work, it is also protecting its own pass-through position. And if you are the prime, the workshare you give away is not just revenue, it is your added-value argument. Keep the design, integration, and acceptance functions in-house even when you subcontract the volume.

ADVANCED

Three Rules, One Question, Different Thresholds

Senior contractors and capture leads should hold all three of these in their heads at once, because a single teaming decision can trip more than one. The FAR pass-through rule asks whether the prime adds value when it subcontracts more than 70% of the work. It applies to covered contracts above a dollar threshold and does not care whether you are a large or small business. The limitations on subcontracting rule (for small business set-asides) asks whether the small business prime is performing enough of the work itself, generally at least 50% of the amount paid for services, with work done by similarly situated small subs counting toward the prime's share. It is about protecting the set-aside, not about markup. ASC 606 asks whether the company controls the service before it transfers to the customer. It does not care about percentages at all; a company that subcontracts 90% can still be a Principal if it genuinely directs and is responsible for the whole. A small business prime that subcontracts 75% of a set-aside services contract to a large business has violated the second rule outright, must disclose under the first, and will probably be an Agent under the third. A large prime that subcontracts 75% of a full-and-open contract while running the integration lab and acceptance testing is fine under all three.

What Acquirers Do With Pass-Through Revenue

When a defense contractor is sold, the buyer's due diligence team separates self-performed revenue from subcontracted revenue and asks about the added value on the latter. Pass-through revenue is typically stripped out of the valuation or valued at a much lower multiple, because it carries thin margin, high concentration risk (the sub can be replaced by a competitor), and possible allowability exposure. A company reporting \$50M of revenue with \$35M of pass-through may be valued as a \$15M company with a \$35M pipe attached. The Revenue Recognition lesson in the Finance module described the acquirer question "are you booking gross or net?" This is the same conversation one level deeper. Sophisticated buyers do not stop at the accounting classification; they ask what functions the company actually performs on subcontracted work and whether those functions would survive the loss of a key subcontractor. Building a business that passes that test is a multi-year effort of moving design, integration, and quality work in-house.

Turning a Pass-Through Contract Into a Value-Add Contract

If you inherit a program that looks like Company A, here is the order of operations. Each step creates evidence for the CO, the auditor, and a future buyer.

Map who actually does what STEP 1

For each subcontracted function, name the prime employee responsible for directing, reviewing, and accepting the work. If the answer is "nobody," you found the problem.

Do this against the SOW paragraphs, not the org chart. The CO and DCAA will compare your claimed functions against SOW tasks and staffing records.

Move acceptance and quality in-house STEP 2

The prime should inspect and accept subcontractor deliverables before they go to the government. This is the cheapest added-value function to establish and the easiest to document.

A written subcontractor quality plan, inspection records, and rejection history are exactly the evidence the FAR definition of added value describes.

Issue technical direction in writing STEP 3

Technical direction letters to subcontractors, within the scope of the subcontract, are proof that the prime directs the work rather than just paying for it.

The MACs and Technical Direction Letters lesson in the Contracting module covers the government-to-prime version. The prime-to-sub version works the same way.

Own the integration and the risk STEP 4

Put the system-level design, integration testing, and warranty on the prime's people. This is what turns an Agent into a Principal.

It also usually improves performance, because someone is finally responsible for the whole thing working.

Fix the SOW and the rates on the next action STEP 5

On the next option, mod, or recompet, make the SOW describe the integration and management work you now do, and price subs with a defensible handling rate.

Do this after the work is real, not before. Describing value you do not yet deliver is the false claims path.

WHAT SENIOR PMS DO DIFFERENTLY

They read their own proposal's pass-through disclosure and make sure the program actually delivers the added value it promised. They keep the evidence file current because they know DCAA will ask years later. They protect the integration, quality, and acceptance functions when the company is looking for places to cut. And they understand that every decision to subcontract a function is also a decision about the company's revenue, its allowable markup, and its price when it is sold.

BUILD ON THIS

- **Revenue Recognition** Principal vs. Agent and the hardware dump problem
- **Subcontractor Management** Managing the sub is the work that makes the markup allowable
- **Wrap Rates & Cost Structure** Where G&A and handling rates come from

Key Terms

Pass-Through

Work a prime contractor buys from a subcontractor and bills to the government with little or no work of its own added. The prime "passes through" the cost, plus indirect cost and profit on top.

Excessive Pass-Through Charges

The government's term for indirect costs and profit a prime applies to subcontracted work when the prime adds no or negligible value. Those charges are unallowable under the FAR limitations on pass-through charges.

Added Value

Subcontract management functions the contracting officer determines benefit the government: technical direction, quality assurance, integration, managing multiple sources, coordinating deliveries, reducing lead times, and similar. The test is what the prime actually does, not what it calls itself.

The 70% Rule

The FAR trigger: if an offeror intends to subcontract more than 70% of the total cost of work, it must disclose that in the proposal and describe the value it adds. The CO then decides whether pass-through charges on that work are allowable.

Principal

Under ASC 606, a company that controls the good or service before it is transferred to the customer. A Principal books the full contract value as revenue (gross).

Agent

Under ASC 606, a company that arranges for another party to provide the good or service without controlling it. An Agent books only its fee or margin as revenue (net).

Subcontract Handling Rate

An indirect rate some contractors apply to subcontract and material costs to recover the cost of managing them, instead of (or in place of part of) G&A. Sometimes called a material handling or value-added rate.

Workshare

The split of work, and therefore revenue and margin, between the prime and its teammates. Negotiated in the teaming agreement and often the most contested number in a partnership.

Self-Performed Work

Work done by the prime's own employees, as opposed to subcontracted. Self-performed work carries the prime's full margin and supports both the added-value and Principal arguments.

Limitations on Subcontracting

A separate rule for small business set-asides requiring the small business prime to perform a minimum share of the work itself (for services, at least 50% of the amount paid). Covered in the Small Business module; mentioned here because it asks a similar question from a different direction.

Knowledge Check

Under the FAR limitations on pass-through charges, what triggers the requirement to disclose and describe added value in a proposal?

A. Any use of subcontractors

✓ **B. Intending to subcontract more than 70% of the total cost of the work**

C. Subcontracting to a foreign company

D. A contract value over \$100M

WHY

The 70% threshold is the trigger. Above it, the offeror must identify the subcontracted amount and percentage and describe the value it adds.

If the contracting officer determines a prime adds no or negligible value to subcontracted work, what happens?

A. The contract is terminated

B. The subcontractor is paid directly by the government

✓ **C. The prime's indirect costs and profit on that subcontracted work become unallowable**

D. The prime must reduce its subcontracting to 50%

WHY

Excessive pass-through charges, meaning the markup applied to work the prime does not meaningfully manage, are unallowable. The prime still pays its subs but does not recover the markup.

Which of these is NOT one of the FAR's examples of added value?

A. Performing quality assurance functions

B. Coordinating deliveries

✓ **C. Holding the prime contract**

D. Managing multiple sources for contract requirements

WHY

Holding the contract is not a subcontract management function. The examples are activities such as QA, coordinating deliveries, managing sources, maintaining inventory, and reducing lead times.

Company A subcontracts \$25M of a \$30M contract and forwards invoices with a markup. An auditor is most likely to classify Company A as:

- A. A Principal booking \$30M
- ✓ **B. An Agent booking only its fee**
- C. A joint venture
- D. A pass-through entity booking \$25M

WHY

Without control of the service (direction, responsibility, risk, pricing), the company is an Agent and recognizes only its fee or margin as revenue.

What is the strongest practical evidence that a prime adds value to subcontracted work?

- A. A paragraph in the proposal describing oversight
- ✓ **B. Written technical direction, QA and acceptance records, and integration test results performed by the prime's staff**
- C. A large G&A rate
- D. The subcontractor's own quality certifications

WHY

Added value is proven by what the prime actually does. Records of direction, inspection, acceptance, and integration are the evidence the CO and DCAA look for.

Why do some contractors use a separate subcontract handling rate instead of full G&A on subcontract costs?

- A. The FAR requires it
- ✓ **B. To price sub-heavy work more competitively and tie the markup to the actual cost of managing subs**
- C. To avoid DCAA audits
- D. To increase profit on subcontracts

WHY

A handling rate built from the cost of procurement, subcontract administration, and QA is both more competitive and easier to defend as added value than a full G&A load.

A small business prime on a services set-aside subcontracts 75% of the work to a large business. Which rule has it violated outright?

A. ASC 606

✓ B. The limitations on subcontracting rule for set-asides

C. The 70% pass-through disclosure rule

D. CAS

WHY

Set-aside primes must generally perform at least 50% of the amount paid for services themselves (or with similarly situated small subs). The 70% rule requires disclosure; the limitations on subcontracting rule is a hard ceiling.

A large prime subcontracts 80% of a full-and-open contract but runs the system design, integration lab, and acceptance testing with its own engineers. Which statement is correct?

A. It violates the pass-through rule because it exceeds 70%

✓ B. It must disclose under the 70% rule, and its documented added value should make the markup allowable and support Principal accounting

C. It is automatically an Agent

D. It cannot charge any profit on subcontracts

WHY

The 70% threshold triggers disclosure, not a prohibition. Real, documented integration and acceptance work demonstrates added value and control.

During due diligence, how do acquirers typically treat pass-through revenue?

A. At the same multiple as self-performed revenue

✓ B. They strip it out or value it at a much lower multiple

C. They ignore it

D. They add a premium for it

WHY

Pass-through revenue carries thin margin, replacement risk, and allowability exposure, so buyers value it far below self-performed revenue.

Match the rule to the question it asks:**CORRECT MATCHES**

FAR limitations on pass-through charges	Does the prime add value when subcontracting over 70%?
Limitations on subcontracting (set-asides)	Does the small business prime perform at least its required share?
ASC 606 Principal vs. Agent	Does the company control the service before it transfers?

WHY

The pass-through rule asks about added value above 70% subcontracting. Limitations on subcontracting asks whether a set-aside prime performs enough itself. ASC 606 asks whether the company controls the service.

Order the steps for converting a pass-through program into a value-add program:**CORRECT ORDER**

1. Map who actually performs each subcontracted function
2. Move acceptance and quality in-house
3. Issue technical direction to subs in writing
4. Own system integration and warranty
5. Update the SOW and handling rates on the next action

WHY

Map who does what, move acceptance and quality in-house, issue written technical direction, own integration and risk, then fix the SOW and rates on the next contract action once the work is real.

A proposal describes an "integration lab" and "supplier quality audits" that the company does not actually perform. Beyond failing the added-value test, what is the main risk?

A. Losing small business status

✓ B. A potential false claims problem, because the government is billed for value not delivered

C. An automatic EVM finding

D. Nothing, as long as the subs perform well

WHY

Charging for management functions that are not performed is misrepresentation, not just an allowability issue. Do the work first, then describe it.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 5

Backlog, Book-to-Bill, and Funded vs. Unfunded

24 min

The three numbers that tell an outsider whether a defense contractor is growing or shrinking, and how a PM's funding mods, option exercises, and descopes move them. Plain-English definitions, a worked example, and how analysts and buyers read the backlog table.

Three Numbers Outsiders Use to Judge the Whole Company

When an investor, a bank, or a potential acquirer looks at a defense contractor, they do not start with the program reviews. They look at three numbers: backlog, funded backlog, and book-to-bill. Together those tell them whether the company's future is bigger or smaller than its present. You already move these numbers. Every funding mod you chase, every option the customer exercises, every task order that gets awarded, and every descope you accept changes them. This lesson explains what they mean so you can see your program the way the people who value the company see it.

Backlog: Revenue That Is Already Won

Backlog is simple in concept. It is the value of work under contract that has not yet been performed. A \$30M three-year contract awarded today adds \$30M to backlog. As the company recognizes revenue over the three years, backlog falls by the same amount. At the end, backlog from that contract is zero and it has all become revenue. Think of it as a bathtub. Bookings are the tap: new awards fill it. Revenue is the drain: work performed empties it. The water level is backlog. If the tap runs faster than the drain, the company is growing.

How Backlog Moves

Ending Backlog = Beginning Backlog + Bookings – Revenue Recognized (± adjustments for descopes, terminations, and EAC changes on cost-type work)

Bookings add. Revenue subtracts. A descope or termination for convenience removes value that was never going to be earned. On cost-type contracts, a funded overrun can actually raise backlog because the contract value grew.

Funded vs. Unfunded: The Distinction That Matters Most

The government awards many contracts, especially cost-type ones, without funding the whole value up front. A \$30M contract might be awarded with \$6M obligated. The full \$30M is backlog. Only \$6M is

funded backlog. The other \$24M is unfunded: real scope, awarded, but the contractor cannot bill it until more money is obligated by a funding modification. Funded backlog is the number that matters for the next twelve months. It is what the company can actually perform and invoice. Unfunded backlog is a promise that depends on the customer's budget arriving on time. The Estimate at Completion lesson in the Finance module covered the Limitation of Funds clause; this is the same idea seen from the company's side. Every funding mod you obtain moves dollars from unfunded to funded, and the finance team notices.

What Counts as Backlog (and What Does Not)

ITEM	IN BACKLOG?	FUNDED OR UNFUNDED?	NOTES
Awarded FFP contract, fully obligated	Yes	Funded	The cleanest case
Awarded CPFF contract, incrementally funded	Yes, full value	Split: obligated portion funded, rest unfunded	Funding mods move dollars from unfunded to funded
Awarded task order under an IDIQ	Yes	Depends on obligation	Task orders are contracts for this purpose
IDIQ ceiling with no orders	No	n/a	Ceiling is potential, not backlog. Never count it.
Unexercised option year	Usually no	n/a	Some companies disclose "priced options" separately
Proposal submitted, not yet awarded	No	n/a	That is pipeline, covered in the Capture module
Protested award	Usually no until resolved	n/a	Conservative companies wait for the protest to clear

Book-to-Bill: Are We Growing or Shrinking?

Book-to-bill is bookings divided by revenue for the same period. If the company booked \$120M of new awards in a year and recognized \$100M of revenue, book-to-bill is 1.2. Backlog grew by \$20M. If it booked \$80M against \$100M of revenue, book-to-bill is 0.8 and the company burned \$20M more backlog than it replaced. Do that for a few years and there is no company left. One quarter can be lumpy: a single large award makes a quarter look like 3.0 and the next look like 0.4. Analysts look at trailing twelve months. A sustained ratio above 1.0 is the definition of a growing contractor.

A Mid-Size Contractor, One Year

\$250M

Beginning backlog
\$140M funded

\$130M

Bookings
Two new awards, three
option exercises

\$110M

Revenue
Backlog burned

1.18

Book-to-bill
Ending backlog \$270M, 2.5
years of revenue

REAL TALK

Every GovCon press release eventually announces record backlog, and every one of them is technically true. Backlog is just the pile of work you are allowed to do someday, and "allowed to" is carrying a lot of weight in that sentence. Before you are impressed, ask two questions: how much of it is funded, and when do the biggest pieces recompute? A \$500M backlog that is 30 percent funded, with its two largest contracts up next spring, is not a \$500M backlog. It is a \$150M backlog wearing a very nice suit.

How Your Everyday Decisions Move These Numbers

- **Chasing a funding mod** Moves dollars from unfunded to funded backlog. No change to total backlog, but a big change to what the company can bill and forecast with confidence.
- **Getting an option exercised** Adds the option value to backlog and counts as a booking. This is often the single largest "booking" a PM produces in a year, and it usually depends on CPARS and customer relationship, not BD.
- **Winning a task order** Adds to backlog and bookings. On a MAC, this is where the real revenue lives; the vehicle award itself added nothing.
- **Accepting a descope** Removes value from backlog that will never become revenue. Sometimes the right call, but report it, because finance has it in the forecast.
- **Growing the EAC on funded cost-type work** Can raise contract value and backlog if the government funds it. Raises revenue too, at lower margin (see the Revenue Recognition II lesson).
- **Losing a recompute** Backlog runs off over the transition period with nothing replacing it. The loss shows up in book-to-bill long before it shows up in revenue.

One sentence to teach your team: **backlog is revenue you have already won, funded backlog is the part you can bill, and book-to-bill above 1.0 is what growth actually looks like.**

INTERMEDIATE

Backlog Quality: Four Questions Behind the Headline Number

Two companies with \$300M of backlog can be worth very different amounts. The difference is quality, and it comes down to four questions. What share is funded? Above 50% is comfortable for a services business; below 30% means the company is exposed to the customer's budget timing. How concentrated is it? If two contracts are 60% of backlog, losing one recompetes is an existential event. When do the recompetes fall? A backlog with most of its value recompeting in the same fiscal year is fragile. What margin is embedded in it? Backlog won at 5% margin to buy market share is worth less than backlog at 12%, and the EAC reviews will surface that later. PMs can answer all four questions for their own contracts. Capture leads and executives roll them up. If your contract is a large share of the company's backlog, the recompetes plan is not a BD exercise, it is the company's survival plan.

Backlog and Revenue Recognition Are Linked

On an over-time contract, backlog and revenue are two views of the same thing: the transaction price minus revenue recognized to date. That means an EAC change moves backlog too. On an FFP contract, backlog is price minus revenue recognized; when the EAC rises and a cumulative catch-up reduces recognized revenue, backlog goes up slightly, which feels wrong but is correct: you have more work left to do for the same money. On a cost-type contract, a funded overrun increases both the transaction price and backlog. The practical point: finance derives backlog from the same contract data you feed the EAC review. If your funded value, ceiling, and remaining scope are wrong in the program's records, the company's reported backlog is wrong too.

Reading Book-to-Bill Together With EAC Adjustments

BOOK-TO-BILL	NET EAC ADJUSTMENTS	WHAT IT USUALLY MEANS
Above 1.0	Favorable	Healthy: winning work and executing it well
Above 1.0	Unfavorable	Growing but struggling to execute; watch margins on new awards
Below 1.0	Favorable	Harvesting: profitable but shrinking; a recompetes or BD problem
Below 1.0	Unfavorable	Trouble on both fronts; expect leadership changes

The IDIQ Ceiling Trap

Small and mid-size contractors routinely present IDIQ ceiling as if it were backlog. "We hold \$2B in contract vehicles" sounds impressive and means almost nothing. A ceiling is a hunting license. Backlog is what you have shot. Sophisticated customers, banks, and buyers know the difference, and a company that

blurs it loses credibility instantly. Report vehicle ceilings as pipeline access, task orders as backlog, and never mix the two in the same number.

ADVANCED

How Acquirers Price Backlog

When a defense contractor is sold, backlog is one of the primary things being bought, and buyers value each layer differently. Funded backlog is valued near face value, discounted only for embedded margin and execution risk. Unfunded backlog is discounted for the probability that funding actually arrives, which depends on the customer, the color of money, and the program's position in the budget. Unexercised options are discounted heavily and often valued on the company's historical option exercise rate. IDIQ ceiling is valued at zero, though a strong vehicle position may add something as "pipeline access." Pipeline (proposals not yet awarded) is valued at Pwin-weighted amounts, if at all. A senior PM preparing a company for sale should therefore prioritize converting unfunded to funded, getting options exercised early, and cleaning up the backlog records so a buyer's diligence team finds exactly what the company claims.

Backlog Conversion and Forecast Accuracy

Finance teams forecast next year's revenue in layers: revenue from funded backlog (high confidence), from unfunded backlog they expect to be funded (medium), from options they expect exercised (medium), and from new awards in the pipeline (low). The share of next year's revenue plan already in funded backlog at the start of the year is called backlog coverage. A services company with 70% coverage is comfortable. One at 40% has to win a lot of work, fast, to make plan. PMs feed this directly. Your funding forecast by month, your option exercise expectations, and your judgment about descopes are the inputs. A PM who says "funding will arrive in Q1" and is wrong by two quarters has just moved revenue out of the year for the whole company. Give finance the realistic date, not the hopeful one.

Backlog Red Flags in Due Diligence

Buyers and lenders look for these. So should you, on your own company.

Ceiling counted as backlog CREDIBILITY

Any backlog number that includes unawarded IDIQ ceiling.

Instant loss of trust. Everything else the company reports gets discounted after this is found.

Funded share below 30% TIMING

The company is exposed to government budget timing it does not control.

Especially concerning if the customer has a history of late funding or the work is on a two-year appropriation nearing expiration.

Top two contracts over 50% of backlog **CONCENTRATION**

One recompetes loss changes the company.

Buyers will price this as a discount or a contingent payment tied to the recompetes outcome.

Backlog growing while margin shrinks **QUALITY**

Winning by underpricing.

Cross-check embedded margin in new awards against the company's historical rate. Backlog bought at 4% margin can be worth less than no backlog.

Recompetes cliff **TIMING**

A large share of backlog recompetes within 18 months.

Look at the incumbent win rate and whether recompetes planning started at award, per the Common Mistakes Contractors Make lesson.

WHAT SENIOR PMS DO DIFFERENTLY

They know their contract's funded and unfunded balance to the dollar and the date the next funding mod is due. They treat option exercise as a deliverable with a plan behind it, not a hope. They report descopes and slips to finance the week they learn of them. And they can explain, in one sentence, how their program moves the company's book-to-bill this year.

BUILD ON THIS

- **Estimate at Completion (EAC)** Ceiling vs funded value and the Limitation of Funds clause
- **The BD-to-Capture Lifecycle** Pipeline is what feeds backlog
- **Common Mistakes Contractors Make** Recompetes planning starts at award

Key Terms

Backlog

The dollar value of work a company has been awarded but has not yet performed and recognized as revenue. It is the company's future revenue that is already under contract.

Funded Backlog

The portion of backlog the government has actually obligated money against. Work the contractor can perform and bill today.

Unfunded Backlog

Contract value that has been awarded but not yet funded. Common on incrementally funded cost-type contracts. Real scope, but the contractor cannot bill it until funds arrive.

Book-to-Bill Ratio

New awards (bookings) in a period divided by revenue recognized in the same period. Above 1.0 means the company added more work than it burned off; below 1.0 means backlog is shrinking.

Bookings

The value of new contract awards, option exercises, and funded task orders recorded in a period. Bookings add to backlog.

Backlog Burn

Revenue recognized in a period, which reduces backlog. A contract enters backlog at award and leaves it as revenue is earned.

Years of Backlog

Total backlog divided by annual revenue. A rough measure of how long the company could operate on work already won. Two to three years is typical for a defense services company.

IDIQ Ceiling

The maximum value that could be ordered under an IDIQ. It is not backlog. Only awarded task orders count. Counting ceiling as backlog is the most common way small contractors overstate their business.

Unexercised Option

A contract option year the government has not yet exercised. Most companies exclude options from backlog until exercised, or report them separately as "priced options."

Backlog Quality

How reliable the backlog is: how much is funded, how concentrated it is in a few contracts, when the big recompetes fall, and what margin is embedded in it.

Knowledge Check

A company starts the year with \$250M backlog, books \$130M, and recognizes \$110M of revenue. What is ending backlog and book-to-bill?

✓ A. \$270M and 1.18

B. \$230M and 0.85

C. \$380M and 1.18

D. \$270M and 0.85

WHY

Ending backlog = \$250M + \$130M - \$110M = \$270M. Book-to-bill = \$130M ÷ \$110M = 1.18.

A \$30M CPFF contract is awarded with \$6M obligated. How does it appear in backlog?

A. \$6M backlog

B. \$30M backlog, all funded

✓ C. \$30M backlog: \$6M funded, \$24M unfunded

D. \$24M backlog

WHY

The full awarded value is backlog. Only the obligated portion is funded; the remainder is unfunded until funding mods arrive.

Which of these should NOT be counted as backlog?

A. An awarded task order

✓ B. An IDIQ ceiling with no orders

C. A funded FFP contract

D. An incrementally funded cost-type contract

WHY

Ceiling is potential, not awarded work. Counting it as backlog is the most common way contractors overstate their business.

Book-to-bill has been 0.8 for three straight years. What is happening?

- A. The company is growing
- ✓ B. Backlog is shrinking; the company is burning more work than it replaces
- C. Margins are improving
- D. The company has too much funded backlog

WHY

Below 1.0 means revenue exceeds new bookings, so backlog falls each year. Sustained, that is a shrinking company.

What does a funding modification on a cost-type contract do to backlog?

- A. Increases total backlog
- B. Decreases total backlog
- ✓ C. Moves dollars from unfunded to funded without changing the total
- D. Counts as new revenue

WHY

Funding mods obligate money against scope already awarded. Total backlog is unchanged; funded backlog rises.

Why is a single quarter's book-to-bill a poor indicator?

- A. Because it is not audited
- ✓ B. Because one large award or a quiet quarter makes it lumpy; analysts use trailing twelve months
- C. Because revenue is only recognized annually
- D. Because it excludes options

WHY

Award timing is uneven. A trailing twelve-month ratio smooths the lumps.

Which combination most clearly signals a company that is "harvesting" rather than growing?

A. Book-to-bill above 1.0 with favorable EAC adjustments

✓ B. Book-to-bill below 1.0 with favorable EAC adjustments

C. Book-to-bill above 1.0 with unfavorable EAC adjustments

D. Book-to-bill below 1.0 with unfavorable EAC adjustments

WHY

Executing profitably (favorable adjustments) while failing to replace backlog (below 1.0) means the company is profitable but shrinking.

How do acquirers typically value unexercised option years?

A. At full face value

B. At zero, always

✓ C. Heavily discounted, often based on the company's historical option exercise rate

D. The same as funded backlog

WHY

Options are not yet contracts. Buyers discount them based on how often the customer has historically exercised.

Order these from most to least reliable as future revenue:

CORRECT ORDER

1. Funded backlog
2. Unfunded backlog
3. Unexercised options
4. Pipeline (proposals submitted)
5. IDIQ ceiling

WHY

Funded backlog is billable now. Unfunded backlog is awarded but depends on funding. Options depend on customer choice. Pipeline is not yet won. Ceiling is only potential.

Match the PM action to its effect:**CORRECT MATCHES**

Funding mod received	Unfunded becomes funded; total unchanged
Option exercised	Booking added; backlog rises
Descope accepted	Backlog reduced; never becomes revenue
Recompete lost	Backlog runs off with no replacement

WHY

Funding mods convert unfunded to funded. Option exercise adds bookings and backlog. Descopes remove backlog. Recompete loss causes backlog to run off.

A company reports \$300M backlog with 25% funded and its two largest contracts (55% of backlog) recompeting next year. What is the best assessment?

A. Strong backlog; nothing to worry about

✓ **B. Low-quality backlog: exposed to funding timing and concentrated in near-term recompetes**

C. The company should count its IDIQ ceilings to improve the picture

D. Backlog quality is irrelevant to valuation

WHY

Low funded share plus heavy concentration in near-term recompetes makes the headline number fragile. Buyers would discount it or tie payment to the recompetes outcome.

What is "backlog coverage" at the start of a year?

A. Total backlog divided by total assets

✓ **B. The share of the year's revenue plan already in funded backlog**

C. The number of contracts in backlog

D. Backlog minus unfunded backlog

WHY

Coverage tells finance how much of the plan is already secured. Low coverage means the company must win a lot of new work in-year to make plan.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 6

Cash Is Not Revenue: How a Defense Contractor Actually Gets Paid

25 min

The gap between doing the work, booking the revenue, sending the invoice, and seeing the money. Vouchers, progress payments, performance-based payments, WAWF, the Prompt Payment Act, fee withholds, and why the PM who bills late is quietly borrowing money for the company at its own expense.

Profitable and Broke at the Same Time

A services contractor wins a \$12M cost-plus contract, staffs it with 40 people, and runs it well. Every month the accountants recognize about \$1M of revenue with a healthy fee. Six months in, the CEO is on the phone with the bank asking to extend the line of credit. How? Because the company has paid twelve payrolls and collected four invoices. Revenue is earned when the work is done. Cash arrives when the invoice is submitted, accepted, and paid. On government contracts the distance between those two events is measured in weeks, and every one of those weeks is financed by the contractor. This lesson walks through how a defense contractor actually gets paid, where the delays come from, and what a PM controls.

IN PLAIN TERMS

Think of a small lawn care business. It pays its crew every Friday, buys fuel every day, and sends customers a bill at the end of the month. Most customers pay a few weeks after that. On paper the business made money in June, but the cash for June's work arrives in August, and the owner covers payroll in between. A defense contractor is in the same spot: revenue is booked when the work is done, but the government's cash comes weeks later.

The Four Events, in Order

On any contract there are four separate events, and they happen in this order: you perform the work, the company recognizes the revenue, the company submits an invoice, and the government pays it. The first two happen the same month. The third depends on your billing cycle and your paperwork. The fourth depends on the government's acceptance and payment process. The What Your CFO Sees lesson introduced DSO, days sales outstanding, as the measure of the gap between invoice and cash. The full gap the company finances is longer than DSO, because it also includes the time between doing the work and invoicing it. If you bill monthly, work done on the first of the month waits 30 days before it is even invoiced.

How Each Contract Type Gets Paid

CONTRACT TYPE	WHAT YOU INVOICE	HOW OFTEN	TYPICAL TIME FROM WORK TO CASH
Cost-type (CPFF, CPAF, CPIF)	Interim public voucher: costs incurred at provisional rates plus fee	Monthly, sometimes twice monthly	45 to 75 days
T&M / Labor-hour	Hours by labor category at contract rates, plus materials	Monthly	45 to 60 days
FFP with progress payments	Progress payment requests (percentage of costs incurred), then delivery invoices	Monthly progress requests; delivery invoices at acceptance	30 to 60 days for progress payments
FFP with performance-based payments	Payment at each defined event or milestone	As events are completed	Depends entirely on event timing and definition
FFP, payment on delivery only	Invoice at delivery and acceptance	At delivery	Entire performance period plus 30 to 45 days

WAWF: Where Invoices Go to Get Paid (or Rejected)

Nearly every DoD invoice is submitted electronically through WAWF. The contractor creates the invoice or voucher, the government acceptor (often your COR) approves it, and the paying office (usually DFAS) pays it. Each of those steps can stall. The invoice can be rejected for a wrong CLIN, a missing receiving report, a mismatch with the contract's funded amount, or a date outside the period of performance. Every rejection restarts the Prompt Payment clock. A voucher rejected twice for paperwork errors can turn a 30-day payment into a 90-day one. The PM does not usually submit the invoice, but the PM is the person who knows whether the CLIN structure, the funding, and the deliverable acceptance are all lined up. The finance team knows how to fill in WAWF. Only the program knows whether what they are filling in matches reality.

The Prompt Payment Act: 30 Days, If the Invoice Is Proper

The government must pay a proper invoice within 30 days of receipt, or it owes interest. The word doing all the work in that sentence is "proper." An invoice missing a required element is not proper, and the clock does not start until it is fixed. Small businesses may be paid faster under accelerated payment policies, but the same properness rule applies. In practice, government interest payments are rare because most delays are caused by improper invoices, not by slow paying offices. The way to get paid in 30 days is to submit invoices that cannot be rejected.

The Cost of the Gap

Annual Financing Cost = (Average Days Work-to-Cash ÷ 365) × Annual Revenue × Borrowing Rate

On a \$12M contract with a 60-day work-to-cash gap and an 8% line of credit: $(60 \div 365) \times \$12M \times 8\% =$ about \$158K per year in interest. And interest is an unallowable cost, so that comes straight out of profit. Cut the gap to 45 days and you save roughly \$40K on this one contract.

INTEREST IS UNALLOWABLE

FAR Part 31 makes interest on borrowings an unallowable cost. The government will not reimburse it on cost-type contracts and will not consider it in pricing fixed-price ones. So every day of delay between doing the work and getting paid is financed by the contractor at its own expense. A PM who lets invoices slip two weeks is not just delaying cash; they are reducing the company's profit by the interest on that cash.

Five Things a PM Controls That Speed Up Cash

- **Clean CLIN and funding alignment** Make sure costs are charged to the right CLIN and that the CLIN has funding before the invoice goes in. A voucher against an unfunded CLIN is rejected automatically.
- **Deliverable acceptance on time** On FFP and milestone-billed work, no acceptance means no invoice. Chase the acceptance the way you would chase a deliverable.
- **Monthly close discipline** Get labor and ODCs posted by the cutoff so the voucher goes out in the first week of the month, not the third.
- **A responsive COR relationship** The COR accepts invoices in WAWF. A COR who trusts your reporting accepts in days; one who does not accepts in weeks.
- **Early warning on funding exhaustion** If the CLIN will run out of money mid-month, the invoice will be partially rejected. Flag it to the CO before the voucher is built.

One sentence to teach your team: **revenue is what we earned, cash is what we collected, and the gap between them is a loan the company gives the government at our expense.**

INTERMEDIATE

Progress Payments vs. Performance-Based Payments

On fixed-price contracts with long performance periods, the government provides financing so the contractor does not have to fund the whole job before delivery. There are two main methods and they

behave very differently for cash. Progress payments pay a percentage of costs incurred, customarily 80% for large businesses and higher for small businesses. They are simple and predictable but leave 20% of costs unfinanced until delivery, and they require an approved accounting system and an audit trail of costs. When you deliver, the delivery invoice is reduced (liquidated) by the progress payments already received. Performance-based payments pay defined amounts when defined events happen: design review complete, first article accepted, test passed. The government prefers them and they can be more cash-favorable than progress payments, sometimes financing up to 90% of contract price, but only if the events are well chosen. A PBP schedule with a big event late in the program is worse for cash than progress payments would have been. The PM should be in the room when the PBP events are negotiated, because the PM is the one who knows which events will actually happen when.

Progress Payments vs. Performance-Based Payments

FEATURE	PROGRESS PAYMENTS	PERFORMANCE-BASED PAYMENTS
Basis	Percentage of costs incurred	Completion of defined events
Typical financing level	80% (large), higher for small business	Up to 90% of price, if events are structured well
Cash predictability	High: tracks spend	Depends on event timing and definition
Admin burden	Requires approved accounting system and cost reporting	Requires event definitions, verification, and a schedule
Risk to contractor	Costs above the percentage are unfinanced	A slipped event delays all cash tied to it
PM leverage	Low: driven by costs	High: PM shapes the event schedule

Provisional Rates and the Closeout Cash Trap

On cost-type contracts you bill indirect costs at provisional billing rates, which are estimates approved for the year. The actual final rates are established later, through the incurred cost submission and DCAA audit, often two to four years after the fiscal year ends. If your final rates come in higher than provisional, the government owes you the difference; lower, and you owe them. Either way, the contract cannot be closed and the final voucher cannot be paid until final rates are settled. Add the fee withhold on CPFF contracts (up to 15% of fee, capped, held until closeout requirements are met) and the picture is clear: a meaningful slice of the cash from a cost-type contract can be stuck for years after the work ends. Companies with a backlog of unclosed contracts are sitting on cash they earned and cannot touch. Getting closeout done is a cash activity, not just a compliance one.

Business System Withholds: When the Government Keeps 5%

DFARS allows the government to withhold a percentage of payments, typically 5% per deficient system, when a contractor's accounting, estimating, purchasing, property, EVM, or material management systems have significant deficiencies. Two deficient systems can mean 10% of every invoice held back until the deficiencies are corrected and verified. The DCAA & DCMA Oversight lesson covers the six business systems; here the point is cash. A system deficiency is not an audit finding to be worked over the next year. It is an immediate reduction in cash on every invoice, on every contract with the clause.

ADVANCED

Cash Forecasting by Contract Mix

A company's cash profile is a function of its contract mix. Cost-type and T&M work converts steadily: bill monthly, collect in 45 to 60 days, with rate settlement risk at the tail. FFP with progress payments converts at 80% of spend with a lump at delivery. FFP with PBPs converts in steps that follow the program schedule. FFP paid on delivery does not convert at all until the end. A company that shifts its mix toward fixed-price development without adjusting its financing is the company that ends up on the phone with the bank. Senior PMs contribute to the company cash forecast directly: expected invoice dates, expected acceptance dates, funding exhaustion dates, and PBP event dates. A program that slips a PBP event by a month has just moved that cash out of the quarter, and the treasurer needs to know in the week the slip becomes likely, not when the invoice fails to go out.

Why Buyers Look at Free Cash Flow, Not Just EBITDA

Earnings measures like EBITDA can be flattered by aggressive revenue recognition and slow collections. Free cash flow cannot. A buyer evaluating a defense contractor will compare cash generated from operations to reported earnings over several years. A company that consistently converts 90% or more of earnings to cash is credible. One that converts 50% has something to explain: growing unbilled receivables, unclosed contracts holding fee, business system withholds, or a habit of booking revenue on unpriced changes that never get paid. The unbilled receivable balance from the Revenue Recognition II lesson is where that explanation usually starts. A rising unbilled balance that is not explained by contract structure (like milestone billing on over-time recognition) is a sign that revenue is being recognized that will not be collected.

Where Cash Gets Stuck on Government Contracts

The common traps, in rough order of how much cash each one holds.

Unbilled work on unpriced changes LARGEST

Work performed under constructive or undefinitized changes that cannot be invoiced until the mod is definitized.

Fix: push definitization hard, document the change file, and consider stopping work if the government will not price it. The Modifications, REAs & Claims lesson covers the mechanics.

Unclosed cost-type contracts YEARS

Fee withholds and final rate settlements waiting on incurred cost audits.

Fix: submit incurred cost proposals on time, push for quick-closeout procedures where eligible, and keep a closeout tracker with the cash value of each open contract.

Rejected vouchers WEEKS

CLIN mismatches, funding shortfalls, missing receiving reports.

Fix: a pre-submission checklist owned jointly by the program and billing, and a monthly look at rejection reasons.

Slow acceptance WEEKS

Deliverables or milestone events completed but not formally accepted.

Fix: treat acceptance as part of the deliverable. A CDRL is not done until the acceptance is in WAWF.

Business system withholds ONGOING

A percentage of every payment held pending system corrections.

Fix: prioritize the corrective action plan like a program milestone. It releases cash on every contract at once.

WHAT SENIOR PMS DO DIFFERENTLY

They know the cash value of their program's unbilled and unpaid balances, not just the revenue. They negotiate PBP events they can actually hit. They treat acceptance, definitization, and closeout as cash milestones. And when they are asked to slip an invoice "to next month" to make a customer's reporting easier, they know exactly what that favor costs the company.

BUILD ON THIS

- **What Your CFO Sees** DSO and the cash gap
- **DCAA & DCMA Oversight** The six business systems that can trigger withholds
- **Modifications, REAs & Claims** Definitizing the changes you cannot bill

Key Terms

Cash Conversion

How quickly and completely recognized revenue turns into cash in the bank. A company can be profitable on paper and still run out of money if conversion is slow.

Public Voucher

The invoice a contractor submits on a cost-reimbursement contract to be paid for costs incurred plus fee. Interim vouchers are submitted during performance; the final voucher comes at closeout.

WAWF / PIEE

Wide Area Workflow, part of the Procurement Integrated Enterprise Environment. The DoD system through which nearly all contractor invoices and receiving reports are submitted, accepted, and paid.

Progress Payments

Payments on a fixed-price contract made before delivery, based on a percentage of costs incurred (customarily 80% for large businesses, higher for small). They are financing, not payment for completed work, and are liquidated against delivery invoices.

Performance-Based Payments (PBPs)

Contract financing on fixed-price contracts tied to measurable events or milestones rather than costs incurred. The government's preferred financing method; often more cash-favorable than progress payments if the events are well defined.

Prompt Payment Act

The law requiring the government to pay a proper invoice within 30 days of receipt (generally) or pay interest. It also defines what "proper" means, which is why rejected invoices restart the clock.

Provisional Billing Rates

Indirect rates approved for interim billing on cost-type contracts before final rates are settled. Set annually; the difference between provisional and final rates is settled at closeout, sometimes years later.

Fee Withhold

On cost-plus-fixed-fee contracts, the government may withhold a portion of fee (up to 15%, capped) until the contractor completes closeout requirements. Cash the company earned but cannot collect until the paperwork is done.

Business System Withhold

Under DFARS, the government may withhold a percentage of payments (typically 5% per system) when a contractor's accounting, estimating, purchasing, or other business systems have significant deficiencies.

Unallowable Interest

Interest on borrowed money is an unallowable cost under FAR Part 31. The government will not reimburse the cost of financing your working capital, so every day of delay in getting paid comes out of profit.

Knowledge Check

A cost-plus contractor has recognized \$6M of revenue but collected \$4M in cash. Which statement is most likely true?

- A. The company is losing money
- ✓ B. The gap is normal: work performed is invoiced monthly and paid 30 to 60 days later
- C. The government has defaulted
- D. Revenue was recognized incorrectly

WHY

On cost-type work there is always a lag between performing work, invoicing it, and receiving payment. The gap is financed by the contractor.

Under the Prompt Payment Act, when does the 30-day payment clock start?

- A. When work is performed
- ✓ B. When a proper invoice is received
- C. When the contract is signed
- D. When the COR is notified

WHY

The clock starts on receipt of a proper invoice. Improper invoices are rejected and the clock restarts when a corrected invoice is received.

Why is a late invoice a profit problem and not just a cash timing problem?

- A. Because the government charges a late fee
- ✓ B. Because interest on borrowed working capital is an unallowable cost the contractor absorbs
- C. Because revenue is reversed
- D. Because DCAA penalizes it

WHY

FAR Part 31 makes interest unallowable. The company finances the gap at its own expense, so every extra day reduces profit.

Progress payments on a large-business FFP contract customarily finance what share of costs incurred?

A. 50%

✓ B. 80%

C. 100%

D. 15%

WHY

Customary progress payments are 80% of costs incurred for large businesses, with higher rates for small businesses. The delivery invoice is later liquidated by the progress payments received.

What is the main cash risk of performance-based payments?

A. They are capped at 50% of price

✓ B. A slipped event delays all the cash tied to it

C. They require an approved accounting system

D. They are only available to small businesses

WHY

PBPs pay on events. If the event slips, the cash slips with it, which is why the PM should shape the event schedule.

A voucher is submitted against a CLIN whose funding has been exhausted. What happens?

A. It is paid from another CLIN

✓ B. It is rejected, and the Prompt Payment clock restarts when a proper invoice is resubmitted

C. It is paid with interest

D. DFAS holds it until the next fiscal year

WHY

Invoices against unfunded amounts are rejected. The PM's job is to flag funding exhaustion before the voucher is built.

Provisional billing rates on a cost-type contract are:

A. The final audited rates

✓ **B. Estimated indirect rates used for interim billing until final rates are settled, often years later**

C. Set by the contractor with no government approval

D. Only used on FFP contracts

WHY

Provisional rates are approved estimates. The difference between provisional and final rates is settled at closeout after the incurred cost audit.

Under DFARS, what can happen to payments when a contractor's business systems have significant deficiencies?

A. Nothing until the next audit

✓ **B. The government may withhold a percentage (typically 5% per system) of each payment**

C. The contract is terminated

D. Fee is doubled to compensate

WHY

Business system withholds reduce cash on every invoice under contracts with the clause until the deficiencies are corrected and verified.

Order the four events from first to last:**CORRECT ORDER**

1. Work is performed
2. Revenue is recognized
3. Invoice or voucher is submitted
4. Government pays

WHY

Work is performed, revenue is recognized in the same period, the invoice is submitted, and the government pays.

Match the payment mechanism to the contract situation:**CORRECT MATCHES**

Interim public voucher	Cost-plus contract, monthly costs plus fee
Progress payment	FFP, percentage of costs incurred before delivery
Performance-based payment	FFP, fixed amounts at defined events
Hours at contract rates	T&M or labor-hour contract

WHY

Cost-type contracts bill interim vouchers. FFP long-duration work uses progress payments or PBPs. T&M bills hours at contract rates.

Which of these is the largest common source of stuck cash on a program?

A. Rounding errors in WAWF

✓ **B. Work performed on unpriced or constructive changes that cannot be invoiced until definitized**

C. Prompt Payment interest

D. Overpaid progress payments

WHY

Undefinitized work is performed at full cost and cannot be billed. Definitization and the change documentation file are cash activities.

Why do acquirers compare free cash flow to EBITDA over several years?

A. To calculate the tax rate

✓ **B. Because earnings can be flattered by aggressive recognition and slow collections, while cash cannot**

C. Because EBITDA excludes government contracts

D. To determine the contract mix

WHY

Consistent conversion of earnings to cash is evidence the reported profit is real. Weak conversion points to unbilled growth, unclosed contracts, or withholds.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 7

Unit Economics of a Billable Hour

26 min

The math of one employee on one contract: bill rate, fully burdened cost, utilization, and the margin left over. Why a five-point drop in utilization can erase a services company's profit, how the bench really gets paid for, and the staffing decisions that quietly set a program's margin.

One Engineer, One Contract, One Number

The Wrap Rates lesson in the Finance module introduced Sarah, an engineer with five layers of cost stacked on her salary. This lesson finishes her story. Sarah's bill rate is set. Her cost is known. Whether the company makes or loses money on her comes down to one more number: how many hours she actually bills. This is the unit economics of a services business. Not the contract, not the program, but the individual billable hour. Every margin problem in a services company can be traced back to it, and every PM sets it, usually without realizing it, through staffing decisions.

IN PLAIN TERMS

Think of a hotel room. The mortgage, cleaning, and utilities cost about the same whether the room is booked or empty, and the nightly rate is already set. The only thing that decides whether the room makes money is how many nights it is actually occupied. A few empty nights a month can wipe out the profit. A billable employee works the same way: salary and burden are fixed, the bill rate is fixed, and utilization (hours actually billed) decides the margin.

The Base Case: Sarah at Full Utilization

Sarah earns \$100,000 a year. Fringe benefits add 30%, overhead adds 20% on top of that, and G&A adds 10% on top of that. Her fully burdened annual cost is roughly \$171,600. On her contract she is billed at \$95 per hour. A year has 2,080 paid hours. Take out ten holidays and three weeks of leave and Sarah has about 1,880 hours she could charge to a contract. If she charges every one of them, she generates $\$95 \times 1,880 = \$178,600$ of revenue. Margin on Sarah: about \$7,000, or 4%. Thin, but positive.

Sarah at 100% Utilization

\$95/hr

Bill rate

Fixed on the T&M contract

1,880

Available hours

After holidays and leave

\$171.6K

Fully burdened cost

\$100K salary plus fringe,
OH, G&A
\$7K (4%)

Margin

Every hour billed

What Happens at 90% Utilization

Now Sarah spends 10% of her available hours on things that are not billable: waiting for a badge, attending an all-hands, filling a gap between task orders. She bills 1,692 hours. Revenue: \$160,740. Her salary and fringe do not change. The company still pays her \$130,000 for the year. Those unbilled 188 hours (about \$12,500 of salary and fringe) move into the overhead pool. On a fixed-rate contract the company cannot pass that on. Sarah's revenue is now \$160,740 against a cost that has not moved. She went from a \$7,000 profit to roughly an \$11,000 loss on the year. A ten-point utilization drop turned a marginally profitable employee into a losing one.

Margin per FTE

$\text{Margin per FTE} = (\text{Bill Rate} \times \text{Available Hours} \times \text{Utilization}) - \text{Fully Burdened Annual Cost}$

Only one term in this formula moves month to month: utilization. Bill rate is contractual. Cost is set by salary and rates. The PM controls utilization through staffing, timing, and task assignment.

Sarah at Different Utilization Levels

UTILIZATION	HOURS BILLED	REVENUE	MARGIN	MARGIN %
100%	1,880	\$178,600	+\$7,000	4%
95%	1,786	\$169,670	-\$1,930	-1%
90%	1,692	\$160,740	-\$10,860	-7%
85%	1,598	\$151,810	-\$19,790	-13%
80%	1,504	\$142,880	-\$28,720	-20%

WHY PROFIT LIVES IN THE LAST 10% OF HOURS

The table above is why services executives obsess over utilization and why "we have a few people on the bench" is never a small problem. The bill rate in this example was deliberately thin, a 1.98 multiplier, which is common on competitive vehicles. On a fatter rate the break-even utilization is lower, but the shape is the same: profit lives in the last 10% of hours.

Where the Bench Cost Actually Goes

When an employee is not billing, the company does not stop paying them. Their time is charged to an indirect account: overhead if they are between assignments, B&P if they are helping write a proposal, G&A

for company-wide activities. Those accounts feed the indirect rates that get applied to everyone's labor. So bench cost does not disappear. It raises the overhead rate, which raises the fully burdened cost of every billable employee, which either erodes margin on fixed-rate contracts or makes the company's proposed rates less competitive. A company with a large bench in the spring shows up with uncompetitive rates in the fall.

How PMs Set Utilization Without Noticing

- **Staffing ahead of funding** Hiring for a task order that starts "next month" and slips two. The new hire sits on overhead while you wait.
- **Holding people through a gap** Keeping a strong engineer on the bench for six weeks between task orders rather than losing them. Sometimes the right call, but it is a real cost that should be a decision, not a drift.
- **Unbilled "extra" work** Letting the team do work the contract does not fund because the customer asked nicely. Every hour is either billable or it is overhead.
- **Slow onboarding** A cleared engineer waiting three weeks for a badge and system access is 3 weeks of overhead. Push the onboarding chain like a deliverable.
- **Meetings and admin on the program's dime** Internal meetings charged to overhead rather than the contract. Legitimate, but it accumulates. Know what share of your team's time is going there.

One sentence to teach your team: **on a services contract the bill rate and the cost are fixed, so the only number that decides whether we make money is how many of the available hours actually get billed.**

INTERMEDIATE

The Multiplier: A Fast Way to Judge a Rate

Divide the bill rate by the employee's base hourly pay. Sarah at \$95 per hour on a \$48 base rate ($\$100K \div 2,080$) is a 1.98 multiplier. Rules of thumb in defense services: below about 1.7 the company is almost certainly losing money at any realistic utilization; 1.8 to 2.1 is competitive and survivable at high utilization; above 2.2 is comfortable and usually only available on less competitive vehicles or highly specialized labor categories. The multiplier is the fastest check a PM can do when asked to staff a labor category with a specific person. If the person's salary implies a multiplier below the company's floor, the answer is either a different person, a different labor category, or a conversation with pricing before the assignment is made.

Salary Creep on Multi-Year Fixed-Rate Work

T&M and IDIQ bill rates are usually fixed for each contract year with a modest escalation, often 2% to 3%. Employees expect raises, and good ones expect more than 3%. On a five-year contract, an engineer hired in

year one at a comfortable 2.0 multiplier can be at 1.75 by year four if raises outpace rate escalation. Multiply across a team and a contract that was profitable at award is break-even by its option years. The defenses: build the staffing pyramid so junior staff grow into mid-level rates over time, rotate people to labor categories that match their new pay, and make sure pricing escalation on the next proposal reflects the company's actual merit increase history rather than a hopeful number.

The Staffing Pyramid: Same Work, Different Margin

TEAM SHAPE	EXAMPLE MIX	BLENDED MULTIPLIER	WHAT IT MEANS
Top-heavy	4 senior, 1 mid, 0 junior	About 1.75	Experienced and expensive; thin or negative margin at fixed rates
Balanced	2 senior, 2 mid, 1 junior	About 1.95	Sustainable margin, room to promote from within
Leveraged	1 senior, 2 mid, 2 junior	About 2.15	Best margin, if the senior lead can carry the quality

The Underrun: Revenue You Were Allowed to Earn

On a T&M contract the government funds a number of hours. If your team bills fewer, the contract underruns. Government PMs sometimes like this; it returns money. For the contractor, every unbilled funded hour is revenue the company had the right to earn and did not, and the unused hours often go back to the customer at year end rather than rolling forward. A program running 8% under its funded hours all year has left 8% of its revenue on the table, usually because positions were unfilled or filled late. Time-to-fill is a revenue metric on a T&M contract.

ADVANCED

The Rate Spiral and How Companies Break It

The mechanism is simple and vicious. Low utilization pushes labor into overhead. Higher overhead raises the fully burdened rate for everyone. Higher rates make proposals less competitive. Fewer wins mean more people on the bench. Lower utilization. Senior leaders at services companies recognize this spiral early because they watch two numbers weekly: company-wide utilization and the gap between actual indirect rates and the provisional rates being billed. Breaking it requires unpopular decisions: releasing bench staff quickly rather than hoping for a task order, freezing hiring until funding is signed, moving indirect work off the overhead pool where legitimate, and in some cases restructuring into separate cost pools so one struggling business unit does not raise rates for the whole company. PMs are usually the first to see the spiral start, because they are the ones holding people through the gap.

Cost-Type Contracts Change the Math (But Not the Discipline)

On cost-reimbursement contracts the company bills actual cost at provisional indirect rates plus fee. If utilization drops and the overhead rate rises, the higher rate gets billed, so the immediate margin damage is smaller than on fixed-rate work. That is why cost-type portfolios feel safer to services executives. Two things keep the discipline in place anyway. First, provisional rates are approved estimates; if actuals run well above them, the company either gets a rate adjustment (with scrutiny) or eats the difference at final settlement. Second, higher rates on cost-type work still make the company less competitive on the next proposal, because evaluators compare proposed rates across bidders. A cost-type contract cushions a utilization problem for a year. It does not solve it.

Pricing Decisions That Set Margin Before the Work Starts

Most margin on a services contract is decided at proposal time. A PM who joins after award inherits these decisions and should understand each one.

Price-to-win vs. price-to-margin STRATEGY

Rates set to beat a competitor rather than to cover cost plus target fee.

Legitimate, if the company knows it and plans the staffing pyramid to make the rates work. Dangerous when the proposal team sets the rates and the delivery team finds out at award.

Escalation assumptions RATES

The annual rate increase built into option years versus actual merit increases.

A 2% escalation against a 4% merit history is a planned margin decline of two points per year.

Labor category mapping STAFFING

Which real people fit which priced categories.

Over-qualified staff in under-priced categories is the most common silent margin killer. The Cost Estimating lesson covers compa-ratio as the check.

Utilization assumption in the rate RATES

Every wrap rate assumes a utilization level. Delivery must hit it.

If pricing assumed 92% and the program runs at 85%, the rate was wrong on day one.

Service Contract Act floors COMPLIANCE

Wage determinations set minimum pay and benefits for covered occupations.

Pricing below the wage determination is not a competitive strategy; it is a violation. Check the WD before pricing any covered services labor category.

WHAT SENIOR PMS DO DIFFERENTLY

They know the utilization assumption baked into their rates and manage to it weekly. They treat open positions on a T&M contract as unearned revenue with a daily cost. They shape the staffing pyramid deliberately and promote from within so pay growth tracks rate growth. And they can tell pricing, before the next proposal, exactly which rates were too thin and why.

BUILD ON THIS

- **Wrap Rates & Cost Structure** How the fully burdened cost is built
- **Cost Estimating** LCATs and compa-ratio
- **Indirect Rate Games** What happens when the overhead pool moves

Key Terms

Bill Rate

The hourly price the government pays for an hour of a given labor category. Fixed on T&M and many IDIQ vehicles; derived from cost plus fee on cost-type work.

Fully Burdened Cost

The total hourly cost of an employee to the company: base pay plus fringe, overhead, and G&A. The wrap rate lesson in the Finance module builds this from the ground up.

Utilization

The share of an employee's available hours that are charged to billable contract work. The single most important number in a services business.

Available Hours

Paid hours minus holidays and leave. A 2,080-hour year typically yields about 1,880 available hours after ten holidays and three weeks of leave.

Bench

Employees who are on payroll but not currently charged to a contract. Their cost goes to overhead, which raises the indirect rates for everyone.

Multiplier

Bill rate divided by the employee's base hourly pay. Defense services multipliers commonly run from about 1.7 to 2.3 depending on the company's cost structure and the vehicle.

Margin per FTE

Annual revenue from one full-time employee minus their fully burdened annual cost. The number that tells you whether adding a person to a contract makes money.

Staffing Pyramid

The mix of senior, mid, and junior staff on a program. A wider base of junior staff, billed at rates that carry a healthy multiplier, usually produces more margin than a top-heavy team.

Underrun

On a T&M or labor-hour contract, billing fewer hours than the contract funds. Every unbilled funded hour is revenue the company was allowed to earn and did not.

Wage Determination

Under the Service Contract Act, minimum wages and benefits the government sets for certain service occupations on covered contracts. It puts a floor under labor cost that pricing must respect.

Knowledge Check

An employee has a fully burdened annual cost of \$171,600, bills at \$95/hour, and has 1,880 available hours. At 100% utilization, what is the annual margin?

✓ A. \$7,000

B. \$17,000

C. -\$7,000

D. \$28,000

WHY

$\$95 \times 1,880 = \$178,600$ revenue. $\$178,600 - \$171,600 = \$7,000$ margin.

Using the same employee, what happens at 90% utilization?

A. Margin rises because overhead falls

✓ B. Revenue drops to about \$160,700 and the employee loses roughly \$11,000 for the year

C. Margin stays at \$7,000 because cost drops proportionally

D. The government pays for the unbilled hours

WHY

Revenue falls to $\$95 \times 1,692 = \$160,740$. Salary and fringe are unchanged, so the employee is now about \$11,000 underwater.

When an employee is on the bench, where does their cost go?

A. It is written off

B. It is billed to the last contract they worked

✓ C. Into an indirect pool (overhead, B&P, or G&A), raising rates for everyone

D. To the government as an ODC

WHY

Unbilled labor charges to indirect accounts, which feed the indirect rates applied to all billable labor.

What is a "multiplier" in services pricing?

A. Utilization times available hours

✓ **B. Bill rate divided by the employee's base hourly pay**

C. G&A rate times overhead rate

D. Fee divided by cost

WHY

The multiplier is a quick check of whether a bill rate can cover a given salary plus burden and still leave margin.

A bill rate implies a 1.6 multiplier on the assigned employee's salary. What does this most likely mean?

A. A healthy margin

✓ **B. The company is probably losing money on that person at any realistic utilization**

C. Nothing; multipliers do not matter on T&M

D. The employee should be promoted

WHY

Below roughly 1.7, the rate does not cover fringe, overhead, and G&A on that salary. The fix is a different person, a different category, or a pricing conversation.

On a five-year T&M contract, rates escalate 2% per year but merit increases average 4%. What is the effect?

A. Margin improves each year

✓ **B. Margin declines about two points per year as pay outpaces rates**

C. No effect; rates and pay are unrelated

D. The government adjusts rates automatically

WHY

Fixed bill rates with faster salary growth compress the multiplier every year. This is salary creep.

Why does a leveraged staffing pyramid (more junior staff) usually produce more margin?

A. Junior staff work more hours

✓ **B. Junior salaries carry a higher multiplier against their labor category rates**

C. Senior staff are unbillable

D. The government pays a premium for junior staff

WHY

Junior categories typically have a wider gap between bill rate and salary, so the blended multiplier is higher, provided the senior lead maintains quality.

A T&M contract funded for 20,000 hours bills only 18,400 hours because two positions were vacant for months. What has happened?

A. The company earned bonus fee

✓ **B. An 8% underrun: revenue the company had the right to earn and did not**

C. Nothing; the hours roll over automatically

D. The government owes the difference

WHY

Unbilled funded hours are lost revenue, and the funding usually does not roll forward. Time-to-fill is a revenue metric on T&M work.

Order the steps of the rate spiral:**CORRECT ORDER**

1. Utilization falls
2. Bench labor charges to overhead
3. Indirect rates rise for everyone
4. Proposed rates become less competitive
5. Fewer wins put more people on the bench

WHY

Low utilization pushes labor into overhead, which raises rates, which makes proposals less competitive, which reduces wins, which lowers utilization further.

Match the PM habit to its effect on unit economics:**CORRECT MATCHES**

Hiring before funding is signed	New hire sits on overhead
Three-week badge delay	Cleared engineer charging overhead
Doing unfunded work the customer asked for	Billable hours converted to overhead
Filling a T&M vacancy in two weeks	Funded hours captured as revenue

WHY

Hiring ahead of funding and slow onboarding create overhead. Unbilled extra work is overhead too. Filling positions fast captures funded hours.

Why does a cost-type contract cushion, but not solve, a utilization problem?

- A. Because the government pays for bench time directly
- ✓ **B. Because higher indirect rates get billed at provisional rates, but actuals above provisional are scrutinized and higher rates hurt future competitiveness**
- C. Because cost-type contracts have no indirect rates
- D. Because utilization does not apply to cost-type work

WHY

The immediate margin hit is smaller on cost-type work, but provisional-to-final settlement and proposal competitiveness still expose the problem.

A wrap rate was priced assuming 92% utilization. The program runs at 85%. What is true?

- A. The rate was too high
- ✓ **B. The rate was wrong from day one for the way the program actually operates; margin will be below plan**
- C. Utilization assumptions do not affect rates
- D. The government will adjust the rate

WHY

Every rate embeds a utilization assumption. Running below it means the fully burdened cost is higher than the rate assumed.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 8

Indirect Rate Games: What Happens to Margin When the Overhead Pool Moves

25 min

Why your program's margin can change without anything on your program changing. Pools and bases, provisional vs. actual vs. final rates, over- and under-absorption, the legitimate moves finance makes and the illegitimate ones CAS forbids, and why a PM's staffing forecast is an input to the whole company's rates.

Your Program Did Nothing Wrong. Your Margin Dropped Anyway.

The quarter closes and your program's margin is a point and a half below plan. You check everything: hours on plan, ODCs on plan, no unbilled work, no EAC change. Then finance explains. The company lost a large contract in another division in March. Forty people went to the bench. The overhead rate went from 20% to 24%. Every hour on every contract, including yours, now costs more to deliver, and your fixed-rate contract cannot pass it on. This is the part of contractor finance that feels the most unfair to PMs and is the least understood. Indirect rates are a company-wide number, set by everyone's behavior, applied to everyone's contracts. This lesson explains the machinery so you can see the moves coming, know which ones are legitimate, and understand why finance keeps asking for your staffing forecast.

Pools and Bases in Sixty Seconds

The Wrap Rates lesson built the stack from the bottom up. Here is the same thing seen from the top. Every cost the company cannot charge directly to a single contract goes into a pool. Fringe (benefits, payroll taxes, leave). Overhead (facilities, program support, bench labor, non-billable management). G&A (executives, finance, HR, business development, legal). Each pool is then spread over a base: the direct costs it is assumed to support. Fringe over labor dollars, overhead over direct labor plus fringe, G&A over total cost input or a value-added base that excludes subcontracts and materials. Rate equals pool divided by base. That one fraction is where every rate game lives. Move the numerator or the denominator and the rate moves for every contract in the company.

IN PLAIN TERMS

Think of four roommates splitting a fixed rent. The rent is the pool, and the roommates are the base. If one roommate moves out, the rent does not drop, so the other three each pay more, even though none of them used any extra space. Indirect rates work the same way: when another division loses work and its direct labor leaves the base, the overhead pool barely changes, so every remaining program carries a higher rate.

The Fraction That Runs the Company

Indirect Rate = Pool Costs ÷ Allocation Base

Example: \$10M overhead pool ÷ \$50M direct labor base = 20%

Lose \$8M of direct labor: \$10M ÷ \$42M = 23.8%

The pool barely moved (the bench went up a little). The base shrank. The rate jumped nearly four points, and every contract in the company now carries it.

Three Rates, Three Times

At any moment a company has three versions of each indirect rate. The provisional rate is the estimate approved before the year started, used to bill cost-type contracts and to price proposals. The actual rate is what the company is really running at, recalculated every month as real pool costs and real base come in. The final rate is what gets settled after the year closes, through the incurred cost submission and DCAA audit, and it is used to true up every cost-type contract retroactively. The gap between provisional and actual is the rate variance, and it shows up on the company's income statement every month as a gain or a loss that no single program owns. When actual runs above provisional, the company is billing less than its true cost on cost-type work and absorbing the excess on fixed-rate work. That is the quiet margin drain.

What a Rate Increase Does to Each Contract Type

CONTRACT TYPE	IMMEDIATE EFFECT	LATER EFFECT
FFP	Margin falls dollar for dollar; nothing can be passed on	Rates on the next proposal are higher, so future wins get harder
T&M / fixed bill rates	Same as FFP: cost up, rate fixed, margin down	Same competitiveness problem
Cost-type, billed at provisional rates	Billing unchanged; company absorbs the gap until rates are adjusted	Final rate settlement recovers the difference, subject to audit and any rate ceilings
Cost-type, provisional rates revised upward mid-year	Billed cost rises, fee in dollars unchanged, margin % falls	Customer notices; scrutiny on the next rate proposal

Under-Absorption: The Company-Wide Margin Miss

Most rate problems are base problems, not pool problems. Companies budget their pools carefully. What they cannot fully control is the base: how much direct labor actually gets charged. Lose a contract, delay a start, run under on hours, and the base shrinks while the pool sits there. The pool is "under-absorbed," the actual rate rises above provisional, and every program in the company reports a margin miss it did nothing to cause. This is why finance asks every PM, every month, for a staffing and hours forecast. Your forecast

is a piece of the base. When ten PMs each forecast slightly optimistic hours, the company sets provisional rates too low, bills too little on cost-type work, and finds out at year end.

YOUR LABOR FORECAST IS THE DENOMINATOR

When finance sends the "please update your direct labor forecast" email, it is not bureaucracy. It is the denominator. A PM who forecasts 40 heads and delivers 34 has, on their own, moved the company's overhead rate. Give the number you actually believe, not the one in the staffing plan.

One sentence to teach your team: **indirect rates are a fraction the whole company shares, and when any division shrinks the base, every program pays the higher rate.**

INTERMEDIATE

Legitimate Moves Finance Makes Mid-Year

When the actual rate drifts above provisional, finance has a handful of proper responses. Cut pool costs: release bench staff, defer discretionary spend, slow indirect hiring. Grow the base: accelerate starts, fill open billable positions, move legitimately direct work off the overhead pool. Revise provisional rates: submit a revised provisional billing rate to the ACO so cost-type contracts bill closer to actual. Restructure pools: split an overhead pool into on-site and off-site pools when customer-site staff use less facility, so the rate more fairly reflects cost. All of these are proper because they change what the company actually spends or how it fairly allocates real cost. What they have in common is that none of them change how a given type of cost is classified from one contract to the next.

The Games CAS Forbids

Cost Accounting Standards 401 and 402 require consistency: a cost of a given type must be treated the same way (direct or indirect) on every contract, every time. That rule closes the doors that would otherwise be tempting. You cannot charge a program manager direct on the cost-type contract and indirect on the FFP one to make the fixed-price margin look better. You cannot move a shared facility cost into a specific contract because that contract has room. You cannot reclassify bench labor as "direct training" on a contract that did not order training. DCAA tests for exactly these patterns in incurred cost audits, and a finding leads to disallowed costs, penalties, and in serious cases a determination that the accounting system is inadequate, which triggers the business system withholds covered in the Cash Is Not Revenue lesson. When a PM is asked to "charge that to the contract" and it does not feel right, the CAS consistency rule is the reason it probably is not.

Proper vs. Improper Rate Responses

MOVE	PROPER?	WHY
Release bench staff to cut the overhead pool	Yes	Reduces real cost
Split overhead into on-site and off-site pools, applied consistently	Yes	Fairer allocation of real cost; must be disclosed and applied to all contracts
Revise provisional billing rates with the ACO	Yes	Brings billing closer to actual; standard process
Charge the same PM direct on one contract and indirect on another	No	CAS 402 consistency violation
Move facility costs into a contract with budget headroom	No	Improper direct charging of an indirect cost
Delay recording pool costs to next year	No	Misstates the rate and the incurred cost submission

Base Choice: Why the G&A Base Matters to Sub-Heavy Programs

G&A can be allocated over total cost input (all direct and overhead costs, including subcontracts and materials) or over a value-added base that excludes subcontracts and materials. The choice changes who pays. Under a total cost input base, a program that is 70% subcontracts carries G&A on all of it, which is exactly the markup the pass-through rule in this module targets. Under a value-added base, subcontracts carry a separate, smaller handling rate and G&A falls only on the company's own effort. Companies choose their base structure once and disclose it; it is not adjusted contract by contract. But PMs on sub-heavy programs should know which structure their company uses, because it determines whether their program is competitive and whether the markup on their subcontracts is defensible.

ADVANCED

The Rate Build: How Next Year's Rates Get Set

Every fall, finance builds next year's indirect budget. It starts with the pool: planned facilities, planned indirect headcount, planned B&P, planned G&A spend. Then the base: every program's direct labor forecast, rolled up. The rate that results goes to the ACO as the proposed provisional billing rate and, for larger contractors, into the forward pricing rate proposal that becomes the FPRA used in every proposal for the year. This is the moment a PM's forecast has the most leverage. A base forecast that is 5% too high produces rates that are too low, which means the company underbills cost-type work all year and prices proposals below true cost. A base forecast that is too conservative produces rates that are too high, which means losing competitions the company could have won. Senior PMs give the rate build their most honest number and flag the risks around it, because the rate that comes out is applied to every hour the company sells.

Rate Ceilings, FPRAs, and Rate Stability as a Competitive Asset

Some contracts, particularly cost-type awards to companies without an FPRA, include indirect rate ceilings in Section H: the government will not reimburse overhead above, say, 25% regardless of actuals. A ceiling turns an under-absorption problem from a settlement issue into a permanent loss. Senior PMs read their Section H for rate ceilings on day one. Companies with an FPRA avoid renegotiating rates on every award and signal to evaluators that their rates are audited and stable. Rate stability itself becomes a discriminator: a customer who has watched a contractor's rates swing five points in two years prices that risk into the evaluation. Predictable rates are the result of predictable bases, which are the result of PMs who forecast honestly and companies that act quickly when the base moves.

Reading a Rate Variance Report

Most contractors circulate a monthly indirect rate report. Here is how a senior PM reads it.

Provisional vs. actual by pool HEADLINE

Which pools are over or under and by how many points.

An overhead variance is usually a base problem (utilization, lost work). A G&A variance is usually a pool problem (spend, B&P). Fringe variance often means benefits costs or leave patterns changed.

Base variance by division CAUSE

Whose direct labor is under forecast.

This is where you find out which division's shortfall is raising your rate. It is also where finance finds out if yours is.

Year-to-date dollar impact MONEY

The rate variance in dollars, split between cost-type (recoverable at settlement) and fixed-rate (absorbed).

The fixed-rate portion is the permanent loss. Watch it grow.

Forecast to year end OUTLOOK

Whether finance expects to recover or whether provisional rates will be revised.

A mid-year provisional rate revision on cost-type work means your customer will see a higher billed rate. Prepare them.

Actions taken RESPONSE

Hiring freezes, bench releases, pool restructuring.

If the actions list is empty two months into a variance, the spiral from the Unit Economics lesson is about to start.

WHAT SENIOR PMS DO DIFFERENTLY

They read the monthly rate report and know which pools are moving and why. They give finance a direct labor forecast they would bet on. They check Section H for rate ceilings before they worry about anything else. They refuse charging instructions that would treat a cost differently from how the company treats it everywhere else. And when their own program is the base problem, they say so before finance finds it.

BUILD ON THIS

- **Wrap Rates & Cost Structure** FPRAs, CAS, and the incurred cost submission
- **Unit Economics of a Billable Hour** Where bench cost comes from
- **DCAA & DCMA Oversight** Who audits the rates and what a finding costs

Key Terms

Indirect Cost Pool

A collection of costs that cannot be charged directly to one contract (facilities, HR, executive salaries, bench labor) grouped for allocation. Common pools: fringe, overhead, G&A.

Allocation Base

The direct cost measure a pool is spread over. Fringe is usually spread over total labor; overhead over direct labor (plus fringe); G&A over total cost input or a value-added base.

Indirect Rate

Pool divided by base. A \$10M overhead pool over a \$50M direct labor base is a 20% overhead rate.

Provisional Rate

The estimated rate approved for billing during the year. Set before the year starts based on the company's budget.

Actual Rate

The rate the company is really running at, recalculated monthly as actual pool costs and actual base come in.

Final Rate

The rate settled after the fiscal year through the incurred cost submission and audit. Used to true up every cost-type contract, sometimes years later.

Under-Absorption

When the actual base is smaller than budgeted, so the pool is spread over fewer dollars and the actual rate is higher than provisional. The most common source of a company-wide margin miss.

Rate Variance

The difference between the provisional rate billed and the actual rate incurred, expressed in dollars. Appears on the company P&L as a gain or loss below program margin.

CAS Consistency

Cost Accounting Standards 401 and 402 require a company to treat like costs the same way (direct or indirect) consistently across all contracts. This is the rule that stops most "rate games."

Forward Pricing Rate Agreement (FPRA)

A negotiated agreement with the government on the indirect rates to use in proposals for a period. Gives pricing stability and reduces negotiation friction on each new award.

Knowledge Check

A \$10M overhead pool is spread over a \$50M direct labor base. The company loses \$8M of direct labor with no change to the pool. What happens to the overhead rate?

A. Falls to 16%

B. Stays at 20%

✓ C. Rises to about 23.8%

D. Rises to 40%

WHY

$\$10\text{M} \div \$42\text{M} = 23.8\%$. A smaller base with the same pool raises the rate for every contract in the company.

What is "under-absorption"?

A. Pool costs are lower than budget

✓ B. The actual base is smaller than budgeted, so the actual rate exceeds provisional

C. Fee is lower than planned

D. DCAA disallowed a cost

WHY

When the base shrinks, the pool is spread over fewer dollars and the actual rate is higher than the provisional rate being billed.

Which contract type absorbs an indirect rate increase immediately with no way to pass it on?

A. CPFF billed at provisional rates

✓ B. FFP

C. CPAF with rate revision

D. None; all contracts pass it on

WHY

Fixed-price and fixed-bill-rate contracts cannot recover higher indirect costs. Margin falls dollar for dollar.

What are the three versions of an indirect rate that exist at any time?

- A. Direct, indirect, and unallowable
- ✓ **B. Provisional, actual, and final**
- C. Fringe, overhead, and G&A
- D. Budgeted, negotiated, and audited

WHY

Provisional is the approved billing estimate, actual is the running rate, and final is the settled rate after audit.

Charging a program manager direct on a cost-type contract and indirect on an FFP contract, to improve the FFP margin, violates:

- A. The Prompt Payment Act
- ✓ **B. CAS 401/402 consistency requirements**
- C. The Anti-Deficiency Act
- D. Nothing; it is a normal practice

WHY

Cost Accounting Standards require like costs to be treated consistently across all contracts. Selective classification is an audit finding.

Which of these is a proper response to an overhead rate running above provisional?

- A. Move facility costs into a contract with budget headroom
- B. Delay recording pool costs until next year
- ✓ **C. Release bench staff and submit revised provisional billing rates to the ACO**
- D. Reclassify bench labor as direct training

WHY

Cutting real pool cost and revising provisional rates through the proper process are legitimate. The other options misstate or misallocate cost.

Why does finance ask every PM for a monthly direct labor forecast?

- A. To calculate award fee
- ✓ **B. Because the forecast is part of the allocation base that sets the company's indirect rates**
- C. To verify timesheets
- D. For CPARS reporting

WHY

Direct labor is the denominator of the overhead rate. Optimistic forecasts produce provisional rates that are too low.

A cost-type contract includes a Section H indirect rate ceiling of 25% on overhead. Actual overhead runs at 28%. What is the consequence?

- A. The government pays 28% at final settlement
- ✓ **B. The 3 points above the ceiling are a permanent loss to the contractor**
- C. The contract converts to FFP
- D. Fee is increased to compensate

WHY

A rate ceiling caps reimbursement regardless of actuals. Costs above the ceiling are not recovered at settlement.

Match the pool to its typical allocation base:**CORRECT MATCHES**

Fringe	Total labor dollars
Overhead	Direct labor plus fringe
G&A	Total cost input or value-added base

WHY

Fringe is spread over labor dollars, overhead over direct labor plus fringe, and G&A over total cost input or a value-added base.

Order the steps of the annual rate build:**CORRECT ORDER**

1. Budget the indirect pools (facilities, indirect staff, B&P, G&A)
2. Roll up every program's direct labor forecast as the base
3. Calculate pool ÷ base for each rate
4. Submit provisional billing rates and the forward pricing rate proposal

WHY

Finance budgets the pools, rolls up every program's direct labor forecast as the base, calculates the rate, and submits it as the provisional billing rate and forward pricing proposal.

Under a total cost input G&A base, a program that is 70% subcontracts:

A. Carries no G&A on the subcontracts

✓ **B. Carries G&A on the full subcontract value, which the pass-through rule may challenge**

C. Is exempt from CAS

D. Gets a lower fee

WHY

TCI bases apply G&A to everything including subs. A value-added base excludes subs and applies a smaller handling rate instead.

On a monthly rate variance report, the fixed-rate portion of the year-to-date dollar impact represents:

A. Money recoverable at final settlement

✓ **B. A permanent loss the company has absorbed**

C. Fee earned

D. Unbilled receivables

WHY

Rate variance on cost-type work is settled later. On fixed-rate work it cannot be passed on and is a realized loss.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 9

How GovCon Companies Are Valued

27 min

What a buyer pays for a defense contractor and why. EBITDA multiples, the six things that move the multiple up or down, the set-aside cliff, novation, and the deal terms that turn a program's weaknesses into money the seller never receives. Written for the PM whose work is quietly setting the price.

The Price Is Set Long Before the Sale

The Revenue Recognition lesson in the Finance module described the acquirer who asks "are you booking gross or net?" in the first meeting. That question is the tip of a much larger process. When a defense contractor is sold, the buyer spends months tearing the business apart, contract by contract, to decide what it is worth. Almost everything they look at was created by program managers, years earlier, doing ordinary program work. This lesson explains what a buyer pays for, what raises or lowers the price, and the deal terms that quietly move money out of the seller's pocket. Even if you never plan to sell a company, the same framework tells you which parts of your program the market considers valuable and which it considers a liability.

Price Equals Earnings Times a Multiple

Most contractor valuations start with a simple formula: adjusted EBITDA times a multiple. EBITDA is the company's operating profit before interest, taxes, and non-cash charges. The multiple is a number that captures how confident the buyer is that those earnings will continue and grow. A services company with \$5M of EBITDA might sell for 7x (\$35M) or 12x (\$60M) depending entirely on what the buyer finds under the hood. The multiple is where all the qualitative judgment lives. Two companies with identical EBITDA can be worth very different amounts, and the difference is mostly about the quality and durability of the revenue behind that EBITDA.

IN PLAIN TERMS

Think of buying a rental house. Two houses bring in the same rent today. One has a tenant on a long lease with a spotless payment record. The other has a tenant whose lease ends in six months and who has paid late before. You would pay more for the first house, because its rent is more likely to keep coming. Buyers value defense contractors the same way: same earnings, but the multiple rises or falls with how durable that revenue looks.

The Starting Point

Enterprise Value $\approx$ Adjusted EBITDA $\times$ Multiple

Then adjust for: net debt, working capital vs. peg, escrows, and earn-outs

The multiplication is the easy part. The adjustments and deal structure are where a \$50M headline becomes \$38M in the seller's account.

What Moves the Multiple

FACTOR	RAISES THE MULTIPLE	LOWERS THE MULTIPLE
Revenue quality	Prime, self-performed, Principal accounting	Subcontract-heavy, pass-through, Agent accounting
Contract mix	Predictable cost-type and long-duration work with healthy fee	Thin-margin T&M or fixed-price development with loss history
Concentration	No contract over 15% of revenue	One contract over 30%, one customer over 50%
Recompete timing	Recompetes spread across years; strong incumbent win rate	Large recompetes in the next 18 months
Small business dependence	Full-and-open wins	Mostly set-aside revenue (the cliff)
Compliance posture	Approved accounting system, current incurred cost years, CMMC in place, clean CPARS	Open audit years, system deficiencies, cyber gaps, mixed CPARS
Growth	Book-to-bill above 1.0, funded backlog rising	Book-to-bill below 1.0
Cash conversion	Free cash flow near EBITDA	Growing unbilled, unclosed contracts, withholds

The Set-Aside Cliff

A large share of small defense contractors are built on set-aside contracts: 8(a), SDVOSB, HUBZone, WOSB, or plain small business. Those contracts are real revenue with real margin. But when the company is bought by a large business, it generally stops being small. The existing set-aside contracts usually continue to completion, but the new owner cannot re-compete them as set-asides. So a buyer treats that revenue as expiring at each contract's end date, not as a going concern. This is the cliff. A \$40M company with \$30M of set-aside revenue is, to a large buyer, a \$10M ongoing business with \$30M running off. Sellers are often shocked by the price. The way around it is to win full-and-open work before the sale, or to

sell to another small business, or to structure the deal so the buyer stays small. All three take years, not months.

EVERY "MAYBE NOT" IS A DISCOUNT

A company is worth what a buyer can keep after the sale. Not what it billed last year. Every contract in backlog gets asked the same question: will this revenue exist in three years under a new owner? Set-asides, key personnel who might leave, a recompetes next spring, and a subcontractor who does all the real work each answer "maybe not," and each "maybe not" is a discount.

What a PM Does That Sets the Price

Almost nothing on the table above is created by the finance team or the owner. CPARS ratings come from program execution. Recompete win rates come from customer relationships and transition planning. Revenue quality comes from how the SOW is written and who does the work. Cash conversion comes from clean invoicing and closeout. Concentration comes from the BD pipeline. A PM who runs a clean, well-documented, high-CPARS program with a real recompetes plan is building the company's sale price whether anyone tells them so or not.

One sentence to teach your team: **a contractor is valued on the earnings a buyer believes will still exist in three years, and every program decision that makes the revenue more durable raises the price.**

INTERMEDIATE

Adjusted EBITDA and the Quality of Earnings Review

Sellers present "adjusted" EBITDA, adding back items they argue are non-recurring: an owner's above-market salary, a one-time legal expense, a bad EAC adjustment they call unusual. Buyers respond with a quality of earnings review, an accounting investigation that tests every adjustment and re-examines the revenue itself. The QoE team re-performs revenue recognition on the largest contracts, checks EACs against actual outcomes, ages the unbilled receivables, and looks for revenue booked on unpriced changes. A PM's EAC history is a direct input. If the last three years show a pattern of EAC increases in the fourth quarter, the QoE report will say the company's earnings are systematically overstated during the year, and the multiple drops. If EACs were conservative and produced favorable adjustments, the report says earnings are of high quality. The Why Q3 Was Bad lesson explained the mechanics; this is where they get priced.

Stock Sale vs. Asset Sale, and Why Novation Matters

In a stock sale the buyer purchases the company itself, so the contracts stay with the same legal entity and generally do not need to be novated. The company may still have to recertify its size status, which is where the set-aside cliff bites. In an asset sale the buyer purchases the contracts and assets, and every government contract must be novated under FAR Subpart 42.12: the contracting officer must agree that the successor can perform and that the government's interests are protected. Novation can take months and is not guaranteed. A CO who is unhappy with the program has leverage at novation. For a PM the message is simple. Your relationship with the CO and your program's performance record become deal risk the moment a sale is contemplated. The buyer's lawyers will ask about both.

Deal Terms That Move Money Away From the Seller

TERM	WHAT IT DOES	WHAT IN THE PROGRAM TRIGGERS IT
Earn-out	Defers part of the price until a result occurs	A large recompetete or option exercise inside the next two years
Escrow / holdback	Holds part of the price against future claims	Open incurred cost years, pending DCAA findings, unresolved REAs
Working capital adjustment	Reduces price if delivered working capital is below the peg	High unbilled receivables, slow collections, large withholds
Key employee retention	Ties price or bonuses to named staff staying	Key personnel clauses; a program that depends on two people
Indemnity for compliance	Seller pays for pre-closing violations found later	Timekeeping, labor mischarging, CAS, export control gaps

Concentration: The Discount Nobody Argues With

Buyers apply the concentration discount almost mechanically. A single contract above about 25% to 30% of revenue, or a single customer above 50%, gets a lower multiple or an earn-out tied to that contract's survival. The logic is unarguable: one contracting officer's decision can remove a third of the business. Diversification is a BD strategy, but it is also a valuation strategy, and it is one reason capture teams pursue new customers even when the existing customer has plenty of work.

ADVANCED

Preparing a Company for Sale: The 24-Month View

Sophisticated sellers start two years before going to market, and most of the preparation is program work. Close out old cost-type contracts and submit every overdue incurred cost proposal so there are no open

audit years. Fix any business system deficiencies and get the accounting system formally approved. Get CMMC and other compliance certifications in place. Convert unfunded backlog to funded and get options exercised early. Move subcontracted functions in-house where the added-value story is weak. Win at least one full-and-open contract if the company is set-aside dependent. Document key processes so the business does not depend on individuals. Clean up revenue recognition so the QoE review finds nothing. None of this is a finance project. Every item is owned by program managers and capture leads. A company whose PMs have been running programs this way all along has nothing to prepare.

Buyer Types and What Each One Pays For

Strategic buyers (other contractors) pay for customer access, contract vehicles, cleared staff, and capabilities that fill a gap. They can often pay more because they will cut duplicate G&A. Private equity buyers pay for predictable EBITDA growth they can lever and later resell; they care most about backlog durability and cash conversion. Small business buyers pay to preserve set-aside revenue and may be the only buyers who can value it at full weight. Knowing which buyer type is in the room tells you which of your program's attributes will be priced.

The Diligence Request List, Program Edition

What a buyer's team will ask the PM for. Having it ready is worth real money.

Contract briefs CONTRACTS

For every contract: type, ceiling, funded value, period of performance, options, recompetes date, set-aside status, key clauses (rate ceilings, key personnel, OCI).

A one-page brief per contract, kept current, is the single most valuable diligence artifact a PM can maintain.

EAC history EARNINGS

Quarterly EACs for the last three years and the reasons for each change.

The QoE team will reconstruct this anyway. Having it ready with explanations shapes the narrative.

CPARS and award fee letters PERFORMANCE

All ratings and determinations, with responses to any below-Satisfactory marks.

Buyers read these as the customer's opinion of the business they are buying.

Staffing and clearances PEOPLE

Roster by contract with clearance level, labor category, tenure, and key personnel designations.

Cleared, tenured staff are an asset. A program that depends on two named individuals is a retention risk priced into the deal.

Subcontractor map REVENUE QUALITY

Who does what on each contract, workshare percentages, added-value documentation.

This is where the pass-through question from earlier in this module gets asked with money attached.

Open items LIABILITIES

Unclosed contracts, open incurred cost years, pending REAs, undefinitized changes, audit findings.

Everything on this list becomes an escrow or an indemnity. Shorter list, higher net price.

WHAT SENIOR PMS DO DIFFERENTLY

They maintain a one-page brief on every contract they run. They know their contract's share of company revenue and the date of its recompute. They keep the added-value documentation and the EAC history current because they know both will be read by strangers someday. And they understand that the difference between a 7x and a 10x multiple is mostly the accumulated quality of program management decisions.

BUILD ON THIS

- **Revenue Recognition** The gross vs. net question the buyer asks first
- **Backlog, Book-to-Bill, and Funded vs. Unfunded** How buyers value each layer of backlog
- **CPAF, Burn Rate & Award Fee** CPARS outlives the contract, and the sale reads it

Key Terms

EBITDA

Earnings before interest, taxes, depreciation, and amortization. A rough measure of operating cash profit used as the base for most contractor valuations.

Multiple

The number a buyer multiplies EBITDA by to arrive at a price. A \$5M EBITDA company at a 9x multiple is worth about \$45M before adjustments.

Adjusted EBITDA

EBITDA with one-time or owner-specific items added back or removed: an owner's above-market salary, a one-off legal cost, a large unfavorable EAC adjustment the seller argues was unusual.

Quality of Earnings (QoE)

A buyer-commissioned accounting review that tests whether reported EBITDA is real, repeatable, and cash-backed. It re-examines revenue recognition, EACs, unbilled balances, and adjustments.

Customer Concentration

The share of revenue from one customer or one contract. High concentration lowers the multiple because one decision by one CO can remove a large piece of the business.

Recompete Risk

The chance that a contract in backlog is lost when it is recompeted. Buyers map every contract's recompete date and incumbent win probability.

Set-Aside Cliff

When a small business is acquired by a large one, it generally loses small business status. Set-aside contracts can continue but cannot be recompeted as set-asides by the new owner, so that revenue is treated as expiring.

Novation

The government's formal recognition of a successor contractor after a sale of assets, under FAR Subpart 42.12. Without it, contracts do not legally transfer. Stock sales usually avoid novation but not recertification.

Earn-Out

A portion of the purchase price paid later only if certain results occur, such as winning a specific recompete or hitting a revenue target. Buyers use earn-outs to shift risk they see in the backlog back to the seller.

Working Capital Peg

The agreed normal level of working capital (receivables, unbilled, payables) that must be delivered at closing. Slow collections or big unbilled balances show up here as price reductions.

Knowledge Check

A contractor has \$5M of adjusted EBITDA and a buyer applies a 9x multiple. What is the approximate enterprise value before adjustments?

A. \$14M

✓ B. \$45M

C. \$50M

D. \$5.9M

WHY

$\$5M \times 9 = \$45M$. Net debt, working capital adjustments, escrows, and earn-outs are then applied.

Which factor most directly **LOWERS** a contractor's valuation multiple?

A. Book-to-bill above 1.0

✓ B. A single contract representing 35% of revenue

C. An approved accounting system

D. Free cash flow close to EBITDA

WHY

High concentration means one CO's decision can remove a large share of the business. Buyers discount for it or tie price to that contract via an earn-out.

What is the "set-aside cliff"?

A. A set-aside contract that exceeds its ceiling

✓ B. When a small business is bought by a large one, set-aside revenue cannot be recomputed as set-aside and is treated as expiring

C. A limit on the number of set-asides a company can hold

D. A DCAA finding on small business status

WHY

Existing set-asides usually run to completion but the new large owner cannot win them again as set-asides, so buyers value that revenue as running off.

In an asset sale, what must happen for government contracts to transfer to the buyer?

A. Nothing; contracts transfer automatically

✓ **B. Novation under FAR Subpart 42.12, with contracting officer agreement**

C. A new competition

D. SBA approval only

WHY

Novation is the government's formal recognition of the successor. It is not guaranteed and gives an unhappy CO leverage.

What does a quality of earnings (QoE) review do?

A. Sets the company's indirect rates

✓ **B. Tests whether reported EBITDA is real, repeatable, and cash-backed by re-examining revenue recognition, EACs, and unbilled balances**

C. Audits the government customer

D. Determines the CPARS rating

WHY

The buyer commissions a QoE to validate earnings. A PM's EAC history is a direct input.

A company shows EAC increases in the fourth quarter three years running. How does a QoE review likely read this?

A. As evidence of strong controls

✓ **B. As a sign earnings are systematically overstated during the year, lowering the multiple**

C. As irrelevant to valuation

D. As a favorable adjustment pattern

WHY

Repeated late-year unfavorable adjustments suggest optimistic interim EACs, so interim earnings are not trustworthy.

Which deal term would a buyer most likely use when a contract representing 30% of revenue recompetes in 14 months?

A. A higher multiple

✓ **B. An earn-out tied to winning the recompute**

C. A working capital peg

D. A stock sale

WHY

Earn-outs shift identified backlog risk back to the seller: part of the price is paid only if the recompute is won.

Large unbilled receivables and slow collections at closing most directly affect the price through:

A. The multiple

✓ **B. The working capital adjustment against the peg**

C. Novation

D. The earn-out

WHY

Working capital below the agreed peg reduces the price at closing. Cash conversion problems land here.

Match the program condition to the deal term it triggers:

CORRECT MATCHES

Open incurred cost years	Escrow or holdback
Large recompute in 18 months	Earn-out
Program depends on two named people	Key employee retention terms
High unbilled receivables	Working capital adjustment

WHY

Open audit years trigger escrows; near-term recompetes trigger earn-outs; two-person dependence triggers retention terms; high unbilled triggers working capital adjustments.

Order these preparation steps by how early they must start before a sale (earliest first):

CORRECT ORDER

1. Win full-and-open contracts to reduce set-aside dependence
2. Move subcontracted functions in-house to strengthen added value
3. Close out old contracts and clear open incurred cost years
4. Assemble contract briefs and EAC histories for diligence

WHY

Winning full-and-open work and moving subcontracted functions in-house take years. Closing audit years and fixing systems take a year or more. Assembling contract briefs and EAC histories is months.

Which buyer type is most likely to value set-aside revenue at full weight?

- A. A large strategic contractor
- B. A private equity fund

✓ C. Another small business that keeps the combined company small

- D. A foreign investor

WHY

Only a buyer that remains small can recompile set-aside contracts, so only that buyer can value the revenue as continuing.

Which of these is created mainly by program management rather than by finance or ownership?

- A. The tax rate

✓ B. CPARS ratings, recompile win rates, and revenue quality

- C. The choice of multiple
- D. The legal form of the sale

WHY

Most factors that move the multiple are the accumulated result of how programs were run: performance, customer relationships, SOW language, and who does the work.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 10

Prime vs. Sub Economics: Who Makes Money on a Team

25 min

The money side of teaming. What a prime earns on a subcontractor's work, what the sub gives up to be on the team, why the government dislikes stacked markups, how joint ventures and contractor team arrangements change the math, and how to decide whether to prime or sub a pursuit.

Every Team Has a Money Story

The Subcontractor Management lesson in the PM Operations module covered how to manage a sub: flow-downs, consent, insight versus oversight. This lesson is about the economics underneath. Who earns what on a team, what each side gives up, and why the government cares about the structure at all. Start with a plain fact: on a teamed contract, the government pays once and two or more companies divide the money. How it is divided is negotiated before award, in a teaming agreement, and fixed after award, in a subcontract. The PM on either side lives with that split for years.

What the Prime Earns on the Sub's Work

When a prime bills the government for a sub's work, the invoice includes the sub's price (which already carries the sub's own fringe, overhead, G&A, and fee) plus the prime's markup: G&A or a subcontract handling rate, plus the prime's fee. On a \$1M subcontract, a prime applying 8% handling and 7% fee bills about \$1.16M. The prime's gross margin on that work is roughly \$160K, of which the handling portion is supposed to cover the actual cost of managing the sub. That is thin compared to self-performed work, where the prime keeps the whole burden stack. The Pass-Through vs. Value-Add lesson explained why: the government pays markup for management, and if there is no management, the markup is unallowable. Primes therefore want subs for what they cannot do themselves, not as a way to grow revenue.

A \$1M Subcontract, From Each Side

\$1.16M

Prime bills government
Sub price + 8% handling +
7% fee

~\$160K

Prime gross margin
Must cover real management
cost

\$1.0M

Sub revenue
With its own full burden and
fee inside

2 layers

Markup the government
pays
Why COs scrutinize teaming

What the Sub Gives Up

The sub keeps its own margin, so on paper it is not losing money by being a sub. What it gives up is everything around the money. It does not get the CPARS record; the prime does. It does not own the

customer relationship; the prime does. It gets paid after the prime gets paid, often 30 days or more later, so it finances the work longer. It performs under flow-downs it did not negotiate. And at recompetes, it is one prime decision away from being replaced. For a small or growing company, the trade is often worth it: revenue, experience, and a foothold with a customer. But it should be a deliberate trade with an exit plan, not a permanent position.

Prime vs. Sub: What Each Side Gets and Gives

DIMENSION	PRIME	SUB
Revenue booked	Full contract value (if Principal)	Own portion only
Margin on the sub's work	Handling markup plus fee	Own full margin
Customer relationship	Owens it	Through the prime
CPARS	Receives the rating	No rating; may cite as past performance
Cash timing	Paid by government in 30 to 60 days	Paid after the prime, often 60 to 90 days
Risk	Liable to the government for everything	Liable to the prime; can be terminated by the prime
Recompete position	Incumbent	Dependent on the prime's team choice

REAL TALK

Being a sub is comfortable in the way a recliner is comfortable. The revenue shows up, the prime handles the customer, and nobody has to write a proposal at eleven at night. Then the prime loses the recompile, or finds someone cheaper, and you discover you spent four years building past performance in somebody else's name. Being a sub is how you get in the door. Priming is how you get to stay in the building.

Why the Government Cares

From the government's side, a stacked markup is money spent on structure rather than work. That is the origin of the pass-through rule, the added-value test, and the limitations on subcontracting for set-asides. It is also why COs ask, at every proposal, who is actually doing the work. A team that is transparent about workshare and can show what the prime contributes gets evaluated on its merits. A team that looks like a shell prime with a working sub gets asked hard questions and often loses on price.

One sentence to teach your team: **the prime earns a thin markup for managing the sub and keeps the customer and the CPARS, while the sub keeps its own margin and gives up almost everything else.**

INTERMEDIATE

Teaming Agreements: Less Binding Than They Look

A teaming agreement is signed before award and typically promises a workshare percentage, exclusivity, and good-faith negotiation of a subcontract if the team wins. Courts in several jurisdictions have treated the workshare and subcontract promises as unenforceable "agreements to agree," because the actual subcontract terms are not yet fixed. The practical result: after award, the prime holds most of the leverage. Workshare "commitments" get renegotiated, scope gets trimmed, and the sub discovers the teaming agreement is a statement of intent. Subs protect themselves by negotiating as much of the subcontract as possible before award (a pre-negotiated subcontract attached to the teaming agreement), by naming specific scope rather than percentages, and by keeping a relationship with the customer that makes them hard to replace. Primes protect themselves by keeping workshare language flexible, because the government's final scope may not match the proposal.

Cash on a Team: Pay-When-Paid and Accelerated Payments

Most subcontracts say the prime pays the sub within some number of days after the prime is paid by the government. If the government takes 45 days and the prime takes another 30, the sub waits 75 days for cash on work it financed from day one. For a small sub, that gap is the difference between making payroll and not. The government has partially intervened. Under FAR 52.232-40, when a prime receives accelerated payments from the government, it must pass them through to small business subs within a short window. Beyond that clause, a sub's cash position is whatever it negotiated. The Cash Is Not Revenue lesson explained why every day of delay is unallowable interest; on a team, the sub is the one paying it.

JV and CTA: Structures That Change the Money

STRUCTURE	WHO HOLDS THE CONTRACT	WHO BILLS THE GOVERNMENT	MARKUP ON TEAMMATES' WORK	WHEN IT IS USED
Prime-sub	Prime	Prime, for everyone	Prime's handling and fee on sub work	Default; most contracts
Joint venture	The JV entity	The JV	None between members; JV overhead instead	Set-aside pursuits, mentor-protégé, large programs
GSA CTA	Each member holds its own Schedule contract	Each member, for its own portion	None	GSA Schedule buys; each firm needs a Schedule

Joint Ventures and the Mentor-Protégé Play

A joint venture forms a new entity that primes the contract. Under SBA mentor-protégé rules, a small protégé and a large mentor can form a JV that qualifies as small for set-asides, provided the protégé manages the JV and performs a required share of the work. For the large mentor, this is access to set-aside revenue it could not otherwise win. For the protégé, it is capability, capital, and a large partner's past performance. The economics: no prime markup between the members, each performs and bills its share through the JV, and profits are split per the JV agreement. The risks are real. Affiliation rules can cost the protégé its small status if the arrangement is not compliant. Workshare requirements are enforced. And when the mentor-protégé agreement ends, so does the JV's eligibility for new set-aside awards. The Small Business module covers the rules; the point here is that the JV changes who earns what.

ADVANCED

Prime or Sub? A Decision Framework

For each pursuit, a company should decide deliberately whether to prime, sub, or JV. The inputs: does the customer know us (if not, priming is hard); do we meet the size and set-aside requirements; can we self-perform enough to satisfy limitations on subcontracting and show added value; do we have the past performance in our own name; can we finance the cash gap on the whole contract, not just our share; do we have the contracts and compliance infrastructure (accounting system, purchasing system, flow-down management) to be the party the government holds accountable. A company that can answer yes to most of these should prime, because the customer relationship, the CPARS, and the incumbent position at recompete are worth more than the sub's margin advantage. A company that cannot should sub, but with named scope, a pre-negotiated subcontract, and a plan to prime the next one.

Negotiating the Split: Where the Leverage Is

Before award, the sub with the scarce capability has leverage: the prime needs its resume, its cleared staff, or its incumbent knowledge. Exclusivity is the currency. A sub that grants exclusivity to one prime should get something binding for it: a pre-negotiated subcontract, named scope, guaranteed key positions. After award, leverage flips to the prime unless the sub controls something the prime cannot replace, usually incumbent personnel the customer wants to keep. Senior contractors on both sides negotiate three things beyond workshare. Rate ceilings on the sub's labor so the prime's pricing holds. Change order pass-through, so a government-directed change flows to the sub with price rather than as free scope. And termination terms, so a prime that loses a task order is not stuck with a sub's minimum commitment.

Team Structures That Get Scrutinized

Patterns that draw attention from COs, evaluators, and later from buyers in diligence.

The shell prime **PASS-THROUGH**

Prime holds the contract; the sub does the work; the prime adds an invoice.

Fails the added-value test, fails Principal accounting, and fails diligence. Covered in the Pass-Through vs. Value-Add lesson.

The 49% large sub on a set-aside **COMPLIANCE**

A small prime with a large sub at the maximum allowable share.

Legal if the small prime really performs its 50%. SBA and COs look at who actually staffs the work, not the proposal percentages.

The ostensible subcontractor **AFFILIATION**

A sub so dominant that SBA treats it as the real prime, making the small prime affiliated and ineligible.

Triggered when the sub performs the primary and vital work, supplies the key staff, or the prime is unusually reliant on it. A size protest on this basis can unwind an award.

Exclusive teaming that blocks competition **COMPETITION**

A prime locking up every capable sub so competitors cannot form a team.

Not illegal in itself, but COs notice, and it invites the government to break the requirement into pieces or restructure the competition.

The sub with the customer relationship **RECOMPETE**

A sub whose staff the customer knows better than the prime's.

The sub is positioned to prime the recompete. Primes that let this happen have outsourced their incumbency.

WHAT SENIOR PMS DO DIFFERENTLY

On the prime side, they keep the customer-facing and integration roles in-house so the prime stays the incumbent. They pass changes to subs with price attached and manage sub cash so a good sub does not fail. On the sub side, they negotiate scope by name, keep a customer relationship the prime cannot cut off, and build past performance in their own name toward the day they prime. On both sides, they know the team's workshare to the percentage point and what it means for margin.

BUILD ON THIS

- **Subcontractor Management** Flow-downs, consent, and managing the sub day to day
- **Pass-Through vs. Value-Add** Why the prime's markup has to be earned
- **The BD-to-Capture Lifecycle** Teaming as a capture decision

Key Terms

Teaming Agreement

A pre-award agreement between a prospective prime and a prospective sub describing roles, workshare, and exclusivity for a specific pursuit. Often not fully enforceable; the subcontract signed after award is what binds.

Subcontract

The post-award contract between prime and sub. It is where price, scope, flow-downs, payment terms, and termination rights are actually fixed.

Workshare

The percentage of contract value (or hours) each team member performs. Determines revenue and margin for each party and drives the prime's added-value and limitations-on-subcontracting positions.

Stacked Markup

The sub's fully burdened price (with its own overhead, G&A, and fee) plus the prime's indirect cost and fee on top. The government pays both layers, which is why it scrutinizes pass-through.

Pay-When-Paid

A subcontract clause under which the prime pays the sub only after the government pays the prime. Shifts cash timing risk to the sub. Limited for small business subs by accelerated payment rules.

Exclusivity

A teaming agreement term restricting a sub from teaming with competing primes on the same pursuit. Valuable to the prime; costly to a sub with scarce capabilities.

Joint Venture (JV)

A separate legal entity formed by two or more companies to bid and perform as the prime. Common in small business mentor-protégé arrangements. Each member performs its share and the JV holds the contract.

Contractor Team Arrangement (CTA)

On GSA Schedules, an arrangement where each team member holds its own Schedule contract and bills the government directly for its portion. No prime markup on teammates' work.

Past Performance Credit

The ability to cite a contract as relevant experience in future proposals. Subs can often cite their portion, but primes get the CPARS record and the customer relationship.

Flow-Down

Prime contract clauses the prime must include in its subcontracts. Covered in the Subcontractor Management lesson; relevant here because flow-downs carry cost and risk to the sub.

Knowledge Check

A prime applies an 8% handling rate and 7% fee to a \$1M subcontract. Approximately what does it bill the government for that work?

- A. \$1.00M
- B. \$1.08M
- ✓ C. \$1.16M
- D. \$1.50M

WHY

$\$1M \times 1.08 \times 1.07 \approx \$1.156M$. The prime's gross margin on the sub's work is about \$160K.

What does a sub typically NOT receive on a teamed contract?

- A. Its own margin on its work
- ✓ B. The CPARS rating
- C. Revenue for its portion
- D. Flow-down clauses

WHY

CPARS goes to the prime. The sub may cite the work as past performance but does not get the rating or the customer relationship.

Why does the government scrutinize stacked markups?

- A. Because subs are not allowed on DoD contracts
- ✓ B. Because it pays two layers of indirect cost and fee, and wants the prime's layer to reflect real management
- C. Because subs cannot hold clearances
- D. Because it prefers JVs

WHY

The pass-through rule and added-value test exist so the prime's markup pays for management, not just for holding the contract.

Why are teaming agreement workshare promises often hard to enforce?

A. Because they are classified

✓ **B. Because courts frequently treat them as unenforceable agreements to agree, since the subcontract terms are not yet fixed**

C. Because the FAR prohibits them

D. Because they expire at award

WHY

The binding document is the post-award subcontract. Subs protect themselves by pre-negotiating it and naming scope.

Under a pay-when-paid clause, when does the sub get paid?

A. When it submits its invoice to the prime

✓ **B. Only after the government pays the prime**

C. Within 7 days regardless

D. At contract closeout

WHY

Pay-when-paid shifts cash timing risk to the sub, which finances the gap. Accelerated payment rules limit this for small business subs.

In a GSA Contractor Team Arrangement, how is the government billed?

A. The prime bills for everyone with markup

✓ **B. Each team member bills directly for its own portion under its own Schedule contract**

C. The JV entity bills

D. GSA bills on behalf of the team

WHY

In a CTA there is no prime markup because each member holds its own contract and invoices its own work.

What is the main advantage of a mentor-protégé JV for the large mentor?

A. Higher fee rates

✓ B. Access to set-aside revenue it could not win alone

C. Exemption from CAS

D. No flow-down obligations

WHY

A compliant mentor-protégé JV can qualify as small, giving the large mentor a share of set-aside work.

Which condition most strongly argues for priming rather than subbing a pursuit?

A. The company has no past performance in its own name

B. The company cannot finance the cash gap on the full contract

✓ C. The company knows the customer, can self-perform most of the work, and has the compliance infrastructure

D. The company lacks an approved accounting system

WHY

Customer knowledge, self-performance, and infrastructure are the preconditions for priming. Without them, subbing with a plan to prime later is the better path.

Match the structure to its markup on teammates' work:**CORRECT MATCHES**

Prime-sub	Prime handling rate plus fee on sub work
Joint venture	None between members; JV overhead instead
GSA CTA	None; each member bills its own portion

WHY

Prime-sub carries the prime's handling and fee. A JV has no markup between members. A CTA has none because each member bills directly.

Order these events in a teaming relationship:**CORRECT ORDER**

1. Sign teaming agreement with workshare and exclusivity
2. Submit the proposal as a team
3. Government awards to the prime
4. Negotiate and sign the subcontract
5. Perform and manage flow-downs
6. Recompete: prime decides whether to keep the sub

WHY

Teaming agreement first, proposal, award, subcontract negotiation, performance, then recompetes.

What is an "ostensible subcontractor" problem?

- A. A sub that is not listed in the proposal
- ✓ **B. A sub so dominant that SBA treats it as the real prime, making the small prime affiliated and ineligible**
- C. A sub without a clearance
- D. A sub paid before the prime

WHY

When the sub performs the primary and vital work or the prime is unusually reliant on it, a size protest can unwind a set-aside award.

A sub with scarce cleared staff is asked for exclusivity by a prime. What should it get in return?

- A. Nothing; exclusivity is standard
- ✓ **B. Something binding: a pre-negotiated subcontract, named scope, or guaranteed key positions**
- C. A higher G&A rate
- D. The CPARS rating

WHY

Exclusivity is the sub's pre-award leverage. Trading it for a non-binding workshare percentage gives it away for free.

THE BUSINESS OF DEFENSE CONTRACTING · LESSON 11

Reading a Prime's 10-K in 20 Minutes

24 min

Everything in this module, applied. A guided path through a public defense contractor's annual report: where the backlog table is, where EAC adjustments hide, how to read the contract-mix disclosure, and what the balance sheet says about cash. Use it on your customer, your prime, your competitor, or the company you are about to join.

Why a PM Should Read One of These

A public defense contractor's annual report is the whole of this module written down by the company itself, with real numbers. Revenue recognition policy, backlog by funded and unfunded, EAC adjustments, contract mix, cash conversion, concentration, and risk. It is free, it is audited, and almost nobody in program management reads it. There are four reasons you should. If you work for a public prime, it tells you what your leadership is measured on. If you sub to one, it tells you whether they are healthy and which segment you are in. If you compete with one, it tells you where their margins are thin. And if you are interviewing with one, reading it is the fastest way to sound like someone who understands the business. This lesson gives you a 20-minute path. No finance background required.

Where to Find It and What Is in It

Search the SEC's EDGAR database for the company name and pick the most recent 10-K. It is long, often 100 to 150 pages, and most of it you will skip. The parts that matter for a PM are five: the Business section at the front, the Risk Factors, the MD&A, the financial statements, and the revenue note in the back. The path below goes through them in the order that builds understanding fastest.

The 20-Minute Path

- **Minutes 0 to 3: Business section, first pages.** Find the segment list and the sentence that says what share of revenue comes from the U.S. government. Note how many segments there are and which one your work falls in.
- **Minutes 3 to 6: Disaggregation of revenue (in the revenue note).** The table splitting revenue by contract type: cost-type versus fixed-price, and by customer. This tells you how much risk the company carries and where.
- **Minutes 6 to 10: MD&A, segment results.** For each segment: revenue, operating profit, margin, and the paragraph explaining the change. This is where net EAC adjustments are usually disclosed. Look for the phrase "net favorable" or "net unfavorable profit adjustments."

- **Minutes 10 to 13: Backlog or RPO.** The backlog table, usually in MD&A: total, funded, unfunded, by segment, and the share expected to convert in the next year. Compute book-to-bill from backlog change plus revenue.
- **Minutes 13 to 16: Balance sheet and cash flow.** Contract assets (unbilled) and contract liabilities. Cash from operations compared to net income. A quick cash conversion check.
- **Minutes 16 to 20: Risk factors, skim.** Search for "fixed-price," "reach-forward," and any program names. This is where the company tells you which programs are hurting.

What Each Section Tells a PM

SECTION	NUMBER TO FIND	WHAT IT MEANS FOR YOU
Business	Percent of revenue from U.S. government	Concentration; how much the company's fate rides on the defense budget
Revenue note	Percent cost-type vs. fixed-price	Fixed-price heavy means EAC discipline matters more; cost-type heavy means rate discipline matters more
MD&A, segments	Segment margin and net EAC adjustments	Which segments are healthy; how much profit depends on risk retirement
MD&A, backlog	Total, funded, next-12-month conversion	Growth or shrinkage; how much of next year is already won
Balance sheet	Contract assets vs. contract liabilities	Whether the company is financing the government or the other way around
Cash flow	Operating cash flow vs. net income	Whether the earnings are real
Risk factors	Named programs and "fixed-price development"	Where the losses are and might be next

A Worked Read-Through: "Prime Co."

Imagine a filing for a company we will call Prime Co. The Business section says 85% of revenue is from the U.S. government across three segments. The revenue note shows 60% cost-type and 40% fixed-price, but one segment, Systems, is 70% fixed-price. In MD&A, Systems shows a 6% operating margin, down from 10%, with net unfavorable profit adjustments of \$180M, and names one fixed-price development program with a reach-forward loss. The other two segments show 11% margins with net favorable adjustments. Backlog is \$40B, 60% funded, with about 35% expected to convert next year. Contract assets grew 20% while revenue grew 5%. Operating cash flow was 80% of net income. In 20 minutes you know: the company is healthy overall, one segment is absorbing a fixed-price development problem, unbilled work is growing faster than revenue (worth watching), and the company has 35% of next year's revenue

already in funded backlog. If you sub to Systems, you now know your prime is under margin pressure and will push on your rates. If you work in Systems, you know why every EAC review is intense right now.

THE HABIT: READ THE PARAGRAPH IN THE 10-Q

The single most useful habit from this lesson: when a large prime announces a charge on a program, open the 10-Q and read the paragraph. It will describe, in the company's own words, exactly the chain of events from the Why Q3 Was Bad lesson: an EAC change, a cumulative catch-up, a segment margin hit. Once you have seen it in a filing, you will never look at your own EAC review the same way.

One sentence to teach your team: **a 10-K is the company's own answer to every question in this module, and reading the segment results, the backlog table, and the revenue note takes twenty minutes.**

INTERMEDIATE

Reading the Revenue Recognition Policy Note

Every filing has a note describing how the company recognizes revenue. It will say that most contracts are recognized over time using a cost-to-cost input method, that award and incentive fees are estimated and included in the transaction price, that changes in estimates are recognized on a cumulative catch-up basis, and that anticipated losses are recognized in full when identified. That is the Revenue Recognition II lesson, verbatim, in the company's own policy. The useful details are the exceptions. Which businesses use point-in-time recognition (usually commercial products or spares)? How does the company treat unexercised options and IDIQs in backlog? Does it disclose gross favorable and gross unfavorable adjustments separately? A company that reports only the net number is making the reader work harder, and analysts notice.

Backlog Table Arithmetic

The backlog disclosure lets you calculate several things the company does not state directly. Book-to-bill: ending backlog minus beginning backlog plus revenue, divided by revenue. Years of backlog: total backlog divided by annual revenue. Funded share: funded divided by total. Near-term coverage: the disclosed percentage expected to be recognized in the next twelve months, multiplied by total backlog, compared to next year's expected revenue. Do this for two or three years of filings and trends appear: a funded share falling from 65% to 50% means the company is winning awards faster than the government is funding them, which is a cash and timing risk. Years of backlog rising while book-to-bill is near 1.0 usually means revenue is flat and awards are long-duration.

Balance Sheet Signals for a Contractor

LINE	HEALTHY PATTERN	WARNING PATTERN
Contract assets (unbilled)	Grows in line with revenue	Grows much faster than revenue: unpriced changes, slow acceptance, or aggressive recognition
Contract liabilities (advances)	Stable; reflects progress or performance-based payments	Sudden drop: financing structure changed, cash will tighten
Receivables (billed)	DSO stable in the 30 to 50 day range	DSO rising: collections or invoice quality problem
Operating cash flow vs. net income	90% or higher over several years	Consistently below 70%: earnings are not converting to cash
Inventories (for hardware companies)	Stable relative to fixed-price production	Rising without revenue growth: production problems or unsold units

Using the Filing Before a Teaming or Rate Negotiation

If you are about to negotiate with a prime, read its most recent 10-Q first. A segment under margin pressure will push subs on rates and workshare; a segment with favorable adjustments and rising backlog has room. If the filing names your program or one like it as a problem, expect scrutiny on every invoice. If contract assets are ballooning, expect slower payment terms. Twenty minutes of reading changes the tone of the negotiation, because you walk in knowing what the other side is measured on.

ADVANCED

Contract Mix Shifts and What They Predict

Track the cost-type versus fixed-price split over three to five years of filings. A company shifting toward fixed-price development is taking on more EAC risk and will show more volatile adjustments in later years; that is often visible in the risk factors before it shows up in the numbers. A company shifting toward cost-type is trading margin ceiling for predictability, and its rate discipline becomes the key variable. For a PM inside the company, the mix shift explains why the EAC process suddenly got more formal or why finance started asking for rate forecasts more often. For a competitor, it shows where the other company is willing to take risk to win.

The Private Company Equivalents

Most defense contractors are private and file nothing public. But the same information exists internally and gets read by the same kinds of people. The bank's covenant package is the private version of the financial statements: trailing EBITDA, leverage ratio, fixed charge coverage. The board deck is the private MD&A:

segment or contract results, backlog, pipeline, EAC adjustments. The quality of earnings report is what a buyer commissions to get the equivalent of an audited 10-K. A senior PM at a private contractor should ask to see the board deck's program pages and the backlog roll-forward; they are the company's own scorecard, and the PM's program is on it.

Questions the Filing Answers About Your Own Program

For a PM at a public prime, the 10-K and 10-Q answer questions about how their program is seen.

How big is my program relative to the segment? SCALE

Segment revenue divided by your annual program revenue.

A 2% program can have a bad quarter quietly. A 15% program cannot. This sets how much scrutiny your EAC gets.

Does my contract type match the segment's risk story? MIX

If the segment is described as shifting to fixed-price, your fixed-price program is part of the narrative.

Expect your EAC to be reviewed by people above your business unit.

Is my program named? EXPOSURE

Search the filing for the program name.

Named programs are watched by analysts. Every subsequent EAC change will be disclosed and explained publicly.

What is the segment's adjustment pattern? CULTURE

Consistent net favorable adjustments suggest conservative EACs are the norm.

A PM who books optimistic EACs in a conservative segment stands out in the wrong way.

How much of next year is funded? PRESSURE

Low near-term coverage means the company needs in-year wins, which means pressure to book awards and exercise options.

Your option exercise and funding mods are part of that plan.

WHAT SENIOR PMS DO DIFFERENTLY

They read their own company's filing every quarter and know the segment margin, backlog, and adjustment story before the all-hands. They read a prime's 10-Q before every teaming or rate negotiation. They read a competitor's to see where it is stretched. And at a private company, they ask for the board deck pages that show their program, because that is the version of the 10-K their leadership is reading.

BUILD ON THIS

- **Why Q3 Was Bad** The EAC adjustment mechanics you will see disclosed
- **Backlog, Book-to-Bill, and Funded vs. Unfunded** The backlog table, explained
- **How GovCon Companies Are Valued** What buyers do with the same numbers

Key Terms

10-K

The annual report a U.S. public company files with the SEC. Contains the business description, risk factors, management discussion, and audited financial statements. Free on the SEC's EDGAR system.

10-Q

The quarterly version of the 10-K, shorter and unaudited. Where the quarterly EAC adjustments and backlog updates appear.

MD&A

Management's Discussion and Analysis. The section where executives explain results in words, by segment. The most readable part of the filing.

Segment

A business unit the company reports separately, such as a division focused on aircraft, space, or services. Segment operating margin is the number analysts compare.

Remaining Performance Obligations (RPO)

The ASC 606 term for backlog: the transaction price allocated to obligations not yet satisfied. Companies disclose it and how much they expect to recognize in the next year.

Disaggregation of Revenue

A required note breaking revenue down by contract type (cost-type vs. fixed-price), customer (U.S. government, international, commercial), and often by segment.

Contract Assets

The balance sheet line for unbilled receivables: revenue recognized but not yet invoiced. Growth here relative to revenue is a cash conversion signal.

Contract Liabilities

The balance sheet line for advance payments and progress billings in excess of revenue recognized. A large balance often means performance-based or progress payments ahead of work.

Net EAC Adjustments

The disclosed total of favorable and unfavorable cumulative catch-ups in the period. Usually in MD&A or the revenue note, sometimes with the gross favorable and unfavorable amounts.

Reach-Forward Loss

The term many filings use for a forward loss on a fixed-price contract, recognized in full when identified. Named individually when material.

Knowledge Check

Where is a public company's annual report available for free?

A. Only from the company's investor relations team

✓ B. The SEC's EDGAR database

C. SAM.gov

D. USASpending.gov

WHY

All 10-K and 10-Q filings are public on EDGAR.

Which section of a 10-K most directly discloses net favorable or unfavorable EAC adjustments?

A. The cover page

✓ B. MD&A segment results (and sometimes the revenue note)

C. The auditor's letter

D. The proxy statement

WHY

Companies explain segment profit changes in MD&A, where net profit adjustments are disclosed.

The disaggregation of revenue note shows a segment is 70% fixed-price. What does that imply?

A. The segment has no EAC risk

✓ B. The segment carries more cost overrun risk, so EAC discipline matters more

C. The segment cannot have unbilled receivables

D. The segment is exempt from ASC 606

WHY

Fixed-price work exposes the contractor to overruns and forward losses. Cost-type work shifts that risk to the government.

Beginning backlog \$38B, ending backlog \$40B, revenue \$30B. What is book-to-bill?

A. 0.93

✓ B. 1.07

C. 1.33

D. 1.25

WHY

Bookings = $\$40B - \$38B + \$30B = \$32B$. Book-to-bill = $\$32B \div \$30B = 1.07$.

Contract assets grew 20% while revenue grew 5%. What is the most likely concern?

A. The company is over-billing

✓ B. Unbilled work is growing faster than revenue: unpriced changes, slow acceptance, or aggressive recognition

C. Backlog is falling

D. Fee is being forfeited

WHY

Contract assets are unbilled receivables. Growth well above revenue growth signals cash conversion risk.

Operating cash flow has been 60% of net income for three years. What does this suggest?

A. Earnings are of high quality

✓ B. Earnings are not converting to cash; something is holding it (unbilled, unclosed contracts, withholds)

C. The company has no debt

D. Backlog is fully funded

WHY

Healthy contractors convert most of their earnings to cash. Persistent low conversion means reported profit is not fully real yet.

What is "remaining performance obligations" in a 10-K?

A. Unresolved DCAA findings

✓ B. The ASC 606 term for backlog: transaction price not yet recognized as revenue

C. Key personnel commitments

D. Undefined contract actions

WHY

RPO is the accounting-standard name for backlog, disclosed with the share expected to convert in the next year.

You are a sub about to negotiate rates with a prime whose latest 10-Q shows the relevant segment with falling margin and a named reach-forward loss. What should you expect?

A. A generous negotiation

✓ B. Pressure on your rates and workshare, and scrutiny on invoices

C. No change; segment results do not affect subs

D. An immediate termination

WHY

A segment under margin pressure pushes cost down through its supply chain. Reading the filing tells you what the other side is measured on.

Order the 20-minute path through a 10-K:**CORRECT ORDER**

1. Business section: segments and share of revenue from the U.S. government
2. Revenue note: cost-type vs. fixed-price split
3. MD&A: segment margins and net EAC adjustments
4. Backlog table: total, funded, near-term conversion
5. Balance sheet and cash flow: contract assets and cash conversion
6. Risk factors: named programs and fixed-price warnings

WHY

Business section, revenue disaggregation, MD&A segment results, backlog, balance sheet and cash flow, then risk factors.

Match the balance sheet line to what it represents:**CORRECT MATCHES**

Contract assets	Revenue recognized but not yet invoiced
Contract liabilities	Billings or advances ahead of revenue recognized
Accounts receivable	Invoiced but not yet collected

WHY

Contract assets are unbilled revenue. Contract liabilities are advance or progress billings ahead of revenue. Receivables are billed and uncollected.

A company's revenue note discloses only net profit adjustments, not gross favorable and unfavorable. Why does this matter?

A. It does not matter

✓ **B. A small net number can hide a large charge offset by many gains; analysts prefer the gross breakdown**

C. It means the company has no fixed-price contracts

D. It violates ASC 606

WHY

Net-only disclosure makes it harder to see problem programs. Gross figures show whether the company is covering losses with gains elsewhere.

At a private contractor, what is the closest equivalent to reading the 10-K?

A. There is none

✓ **B. The bank covenant package and the board deck's program and backlog pages**

C. SAM.gov entity registration

D. The CPARS report

WHY

Private companies produce the same information for lenders and boards. A senior PM should ask to see the pages that show their program.

THE BUSINESS OF DEFENSE CONTRACTING · MODULE ASSESSMENT

Module Assessment

Pass this to unlock the next skill level in the app.

Knowledge Check

On an over-time contract using the cost-to-cost method, which number is the denominator of the percent-complete calculation?

- A. Contract price
- B. Funded value

✓ C. Estimated cost at completion (EAC)

- D. Cumulative billings

WHY

Percent complete equals costs incurred to date divided by the estimated cost at completion. The PM's EAC directly drives recognized revenue.

When the EAC on a fixed-price contract exceeds the contract price, the company must:

- A. Spread the loss over the remaining work

✓ B. Recognize the entire projected loss immediately

- C. Stop recognizing revenue
- D. Request a contract modification

WHY

Forward losses are recognized in full in the period they are identified, regardless of percent complete.

Why does a mid-program EAC increase produce a profit hit larger than the current period's share of the overrun?

- A. Because DCAA adds penalties

✓ B. Because the cumulative catch-up reprices all work already performed

- C. Because revenue is deferred
- D. Because fee is forfeited

WHY

The catch-up brings profit-to-date in line with the new estimate all at once, reversing profit previously recognized on past work.

Under the FAR limitations on pass-through charges, an offeror must describe its added value when it plans to subcontract more than what share of the total cost of work?

A. 50%

B. 60%

✓ C. 70%

D. 80%

WHY

The 70% threshold triggers disclosure of the subcontracted amount and a description of the added value the prime provides.

A company begins the year with \$250M backlog, books \$130M, and recognizes \$110M of revenue. Book-to-bill is:

A. 0.85

✓ B. 1.18

C. 2.27

D. 0.44

WHY

Book-to-bill = bookings ÷ revenue = \$130M ÷ \$110M = 1.18. Ending backlog is \$270M.

Why is a late invoice a profit problem and not only a timing problem?

A. The government charges a fee

✓ B. Interest on borrowed working capital is unallowable, so the contractor absorbs the financing cost

C. Revenue is reversed

D. It triggers a business system withhold

WHY

FAR Part 31 makes interest unallowable. Every day between work and cash is financed at the contractor's expense.

An employee bills at \$95/hour with 1,880 available hours and a fully burdened cost of \$171,600. At 90% utilization the result is approximately:

A. A \$7,000 profit

✓ B. An \$11,000 loss

C. Break-even

D. A \$17,000 profit

WHY

Revenue at 90% is $95 \times 1,692 = \$160,740$, against an unchanged cost of \$171,600. Profit lives in the last 10% of hours.

A \$10M overhead pool over a \$50M direct labor base gives a 20% rate. The base falls to \$42M with no pool change. The rate becomes about:

A. 16%

B. 20%

✓ C. 23.8%

D. 30%

WHY

$\$10M \div \$42M = 23.8\%$. A shrinking base raises the rate for every contract in the company.

What is the "set-aside cliff" in a company sale?

A. A limit on set-aside contract value

✓ B. When a small business is bought by a large one, set-aside revenue cannot be recomputed as set-aside and is treated as expiring

C. A DCAA finding on size status

D. A cap on the purchase price

WHY

Buyers value set-aside revenue as running off unless the buyer stays small, which is why full-and-open wins raise the multiple.

A prime applies 8% handling and 7% fee to a \$1M subcontract. What does the sub give up that the prime keeps?

A. Its own margin

✓ **B. The CPARS rating and the customer relationship**

C. Revenue for its portion

D. Its indirect rates

WHY

The sub keeps its margin on its work but the prime holds the CPARS record, the customer relationship, and the incumbent position at recompete.

In a 10-K, contract assets growing 20% while revenue grows 5% most likely signals:

A. Strong cash conversion

✓ **B. Unbilled work growing faster than revenue: unpriced changes, slow acceptance, or aggressive recognition**

C. Falling backlog

D. Higher fee

WHY

Contract assets are unbilled receivables. Growth far above revenue growth is a cash conversion warning.

A company that controls a service before it transfers to the government, directs the subcontractors, and bears performance risk is, under ASC 606:

A. An Agent booking net revenue

✓ **B. A Principal booking gross revenue**

C. A pass-through entity

D. A joint venture partner

WHY

Control before transfer is the Principal test. Principals recognize the full contract value as revenue.